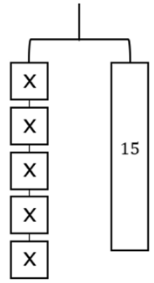
Unit 14 Overview: Equations and Inequalities

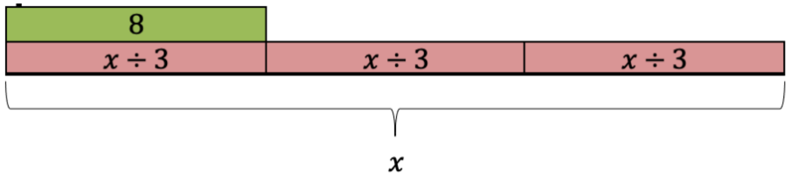
Unit Narrative

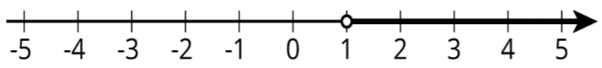
In Unit 14, students build on the knowledge acquired in Unit 13 to solve one-step equations and inequalities. The unit also includes representing real-world situations by equations and inequalities. In this unit, students transition from using mental math and visual models to using inverse operations to solve equations algebraically. Throughout the unit, students make connections between written descriptions and algebraic expressions.



In Section 1 of this unit, students apply their knowledge from Unit 13 to represent real-world scenarios with inequalities. They also solve one-step rational-number equations using mental math and visual models. The equations students encounter in Lesson 2 include all four of the basic operations, relate three numbers, and have the unknowns on either side of the equal sign in any position.

Section 2 extends concepts of solving equations using mental math and visual models to use inverse operations to solve equations algebraically. Initially, students develop a conceptual understanding of inverse operations. Then students apply their knowledge of inverse operations to solve one-step equations, many of which represent real-world situations.





In Section 3, students return to inequalities to explore solving one-step inequalities and representing the solutions on a number line. Previously, in Unit 13, students used substitution to check if solutions satisfied inequalities. In this unit, students apply what they have learned about solving one-step equations to solve one-step inequalities using the properties of inequalities. Students find the minimum/maximum values for an inequality and interpret which solutions make sense in the given contexts. Students also discover why you reverse the direction of the inequality symbol when you multiply or divide the inequality by a negative number.

Unit At-A-Glance

Section	Benchmarks Addressed	Lesson	Title
Section 1: Exploring Equations and Inequalities	**MA.6.AR.1.2** **MA.6.AR.2.4**	Lesson 1	Translating Real-World Written Descriptions Into Algebraic Inequalities
		Lesson 2	Finding an Unknown Value in an Equation
Section 2: Writing and Solving One-Step Equations	**MA.6.AR.2.2** **MA.6.AR.2.3**	Lesson 3	Discovering Inverse Operations
		Lesson 4	Solving One-Step Equations in One Variable Using Addition and Subtraction
		Lesson 5	Writing and Solving One-Step Addition and Subtraction Equations Within Real-World Contexts
		Lesson 6	Solving One-Step Multiplication and Division Equations in One Variable Using Models
		Lesson 7	Solving One-Step Multiplication and Division Equations Algebraically
		Lesson 8	Writing and Solving One-Step Equations in One Variable
Section 3: Solving One-Step Inequalities	**MA.7.AR.2.1**	Lesson 9	Solving One-Step Inequalities Using Addition and Subtraction
		Lesson 10	Solving One-Step Inequalities Using Multiplication and Division

Differentiation

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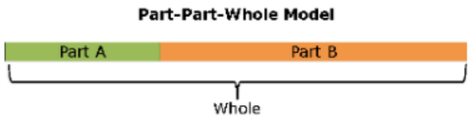
Engagement
The "WHY" of learning

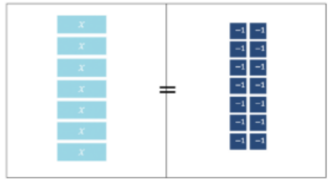
- Lessons throughout this unit provide opportunities for students to develop an appreciation for the utility of mathematics as they make connections between real-world written descriptions and algebraic expressions, equations, and inequalities.
- The Wrap-Up components in multiple lessons provide students with the opportunity to self-regulate their learning as they self-assess their understanding of learning targets.

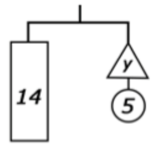
Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Representation
The "WHAT" of learning

- Students engage with kinesthetic and visual entry points through their use of concrete manipulatives, such as algebra tiles, and visual models, such as bar models, to model algebraic expressions and equations. As the unit progresses, students continue to build on their understanding of algebraic expressions and use another visual model, hanger diagrams, to visually represent balancing equations. The use of algebra tiles and the visual models provides necessary scaffolding for students to solve one-step equations and inequalities algebraically without the added support of diagrams and manipulatives by the end of the unit.
- Properties of equality and inequality are used throughout this unit. Consider posting the properties of equality and properties of inequality on an anchor chart in your room for students to reference. Consider also providing visual examples of each property.

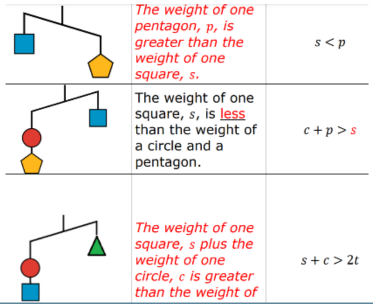






Action & Expression
The "HOW" of learning

- Students have opportunities to respond in a variety of ways, including written descriptions, algebraically, and verbally, including the use of sketches or manipulative tools like algebra tiles or bar models.
- Provide students with access to assistive technologies, such as audio recorders, speech-to-text software, and virtual manipulatives.
- The Lesson 10 Warm-Up provides a visual entry-point as students connect hanger diagrams to inequalities.



ELL Information

Mark Up Text

In many lessons throughout the unit, students will be solving real-world problems. This topic is text and reading heavy. When students translate words into numbers, algebraic expressions, or equations, encourage them to use a highlighter to highlight phrases and words to call attention to action words and the action of a story in real-world contexts.

The teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers and how to write those numbers in an expression or equation.

The teacher will provide students with the following symbols:

- ✓ - background knowledge; they know what to do.
- ? - confusing information; is this needed to solve the problem?
- ! - marks where a variable is needed to represent a number.
- + - extra info in word problem; not needed.

Using a small piece of text, the teacher will model for students how you would mark up before releasing them to do the same. Students will read the text slowly and carefully, focusing on one sentence at a time and marking text based on the key. Then they will turn to a partner to discuss each sentence and compare where they made marks. For the first few examples, have groups share their decisions out to the class. Then guide students through turning the marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support."

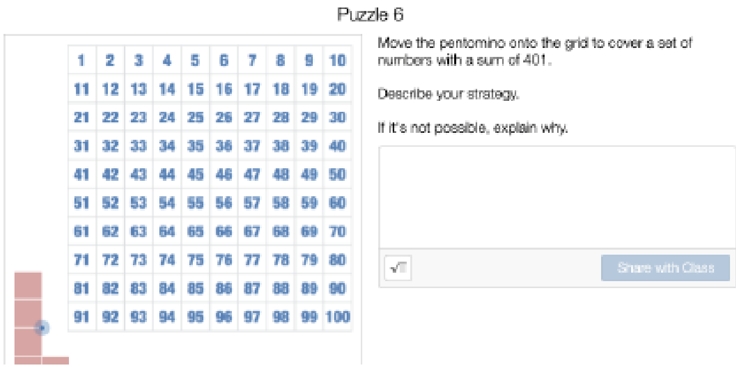
- Equation
- Asymmetric property of inequality
- Addition property of inequality
- Multiplication property of inequality
- Subtraction property of inequality
- Division property of inequality

Additional Differentiated Support

Scaffolding Resources

Pentomino Puzzles

The link to this activity can be found in the Teacher Area. This Desmos Activity requires students to work through a series of pentomino sum puzzles. The activity begins with students working to find a sum of 65 and progresses as students use algebraic reasoning to solve puzzles and create algebraic expressions to represent them. This activity encourages students to reason about the efficiency of algebraic expressions and promotes mathematical justification as students strategize to complete each puzzle.



On-Ramp

On-Ramp to 6th Grade provides video and practice tools for students to scaffold the following skills:

- Determine the Unknown Whole Number in a Multiplication Equation
- Determine the Unknown Whole Number in a Division Equation
- Adding Decimals
- Subtracting Decimals
- Multiplying Decimals
- Dividing Decimals
- Adding Two-Digit and One-Digit Numbers
- Addition Fluency
- Subtraction Fluency

Section Overview

Section 1: Exploring Equations and Inequalities (Lessons 1-2)

Benchmarks Addressed	Learning Targets	
MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. • *Clarification 1: Variables may be on the left or right side of the inequality symbol.*	Lesson 1	• I can translate a written description into an algebraic inequality. • I can represent an inequality on a number line when given a written description of a real-world context.
MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position. • *Clarification 1: Instruction focuses on using algebraic reasoning, drawings, and mental math to determine unknowns.* • *Clarification 2: Problems include the unknown and different operations on either side of the equal sign. All terms and solutions are limited to positive rational numbers.*	Lesson 2	• I can determine an unknown decimal in a numeric equation with three numbers. • I can determine an unknown fraction in a numeric equation with three numbers.

Section 2: Writing and Solving One-Step Equations (Lessons 3-8)

Benchmarks Addressed	Learning Targets	
MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. • *Clarification 1: Instruction includes using manipulatives, drawings, number lines and inverse operations.* • *Clarification 2: Instruction includes equations in the forms $x + p = q$ and $p + x = q$, where x, p and q are any integer.* • *Clarification 3: Problems include equations where the variable may be on either side of the equal sign.*	Lesson 3	• I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations. • I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.
	Lesson 4	• I can solve addition and subtraction one-step equations.
	Lesson 5	• I can write and solve one-step equations in one variable within a real-world context using addition and subtraction.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers. • *Clarification 1: Instruction includes using manipulatives, drawings, number lines and inverse operations.* • *Clarification 2: Instruction includes equations in the forms $\frac{x}{p} = q$, where $p \neq 0$, and $px = q$.* • *Clarification 3: Problems include equations where the variable may be on either side of the equal sign.*	Lesson 6	• I can solve one-step multiplication and division equations using manipulatives. • I can solve one-step multiplication and division equations using diagrams.
	Lesson 7	• I can solve one-step multiplication and division equations in one-variable algebraically.
	Lesson 8	• I can write and solve one-step equations in one variable within a real-world context using multiplication and division. • I can solve one-step equations in one variable using addition and subtraction.

Section 3: Solving One-Step Inequalities (Lessons 9-10)

Benchmarks Addressed		Learning Targets
MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically. • *Clarification 1: Instruction focuses on the properties of inequality. Refer to Properties of Operations, Equality and Inequality (Appendix D).* • *Clarification 2: Instruction includes inequalities in the forms $px > q$, $\frac{x}{p} > q$, $x \pm p > q$ and $p \pm x > q$, where p and q are specific rational numbers and any inequality symbol can be represented.* • *Clarification 3: Problems include inequalities where the variable may be on either side of the inequality symbol.*	Lesson 9	• I can relate the solution set of an inequality to its graph on a number line. • I can relate the solution set of an inequality to the context it represents. • I can determine the minimum or maximum value that makes an inequality true. • I can solve one-step inequalities in one variable using addition or subtraction. • I can relate properties of inequality to solving one-step inequalities algebraically.
	Lesson 10	• I can determine the minimum or maximum value that makes an inequality true. • I can explore the relationship between the coefficients of the terms in an inequality and solution set. • I can solve one-step inequalities in one variable using multiplication or division. • I can relate properties of inequality to solving one-step inequalities algebraically.

Vertical Alignment

Grade 6 Accelerated Unit 14
Equations and Inequalities

→

Grade 7 Accelerated Unit 1
Equations and Inequalities

Prior Course(s)	Course Level	Future Course(s)
MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions. **MA.5.AR.2.3** Determine and explain whether an equation involving any of the four operations is true or false. **MA.5.AR.2.4** Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	**MA.6.AR.1.2** Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. **MA.6.AR.2.2** Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. **MA.6.AR.2.3** Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers. **MA.6.AR.2.4** Determine the unknown decimal or fraction in an equation involving any of the four operations relating three numbers with the unknown in any position. **MA.7.AR.2.1** Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.	**MA.7.AR.2.2** Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers. **MA.8.AR.2.2** Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

14.1 Translating Real-World Written Descriptions Into Algebraic Inequalities

Lesson Introduction

Benchmark	MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.	Learning Targets	• I can translate a written description into an algebraic inequality. • I can represent an inequality on a number line when given a written description of a real-world context.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Essential Questions

- How do you write an inequality from a verbal description?
- How do you represent inequalities on a number line?
- Why is the real-world context important when determining possible solutions to inequalities?

Lesson Narrative

Students translate verbal descriptions representing a real-world situation into an inequality that represents the situation. They also represent inequalities on number lines. Students use the written description to determine which symbol to use and how it should be represented on a number line.

Component Summary

14.1.1 Warm-Up (~5 min.)	Estimate how many TVs it will take to cover a jumbotron *(SE p. 335)*	14.1.4 Try It (5-10 min.)	Practice translating verbal descriptions of real-world situations into inequalities and represent them on a number line *(SE p. 339)*
14.1.2 Exploration (10-15 min.)	Explore how to represent inequalities on a number line *(SE p. 336)*	14.1.5 Your Turn! (5 min.)	Represent two inequalities on number lines and determine possible values *(SE p. 340)*
14.1.3 Guided Practice (10 min.)	Translate verbal descriptions of real-world situations into inequalities and represent them on a number line *(SE p. 338)*	14.1.6 Wrap-Up (~5 min.)	Demonstrate understanding of writing an inequality, determining possible values that satisfy the inequality, and representing the inequality on a number line *(SE p. 341)*

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> N/A <u>Additional Terms</u>: • Inequality • Strict inequality	N/A	• (Optional) Locate pictures and videos of the real-world contexts presented in each section. • (Optional) Gather information about students' interests to substitute in for some of the real-world contexts.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may struggle to interpret inequalities when the variable is on the right.
Students may struggle to determine the minimum and maximum value of the inequality.
Students may struggle translating verbal descriptions when they involve greater than or equal to and less than or equal to, because these are new in this lesson.</td><td>Consider modeling how to read the $\leq$ and $\geq$ symbols at the beginning of the lesson for students. This will be particularly helpful for ELL students.
Discuss “at least” and “at most” to make sure all students understand what they mean in various contexts.
Consider showing pictures or videos to help students understand the context of the problems. For example, in 14.1.3 Guided Practice, show an illustration of the lines on a highway.
If time is an issue, consider assigning 14.1.5 Your Turn! for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
In 14.1.2 Exploration, have students first complete the questions independently and then discuss their work with a partner. These discussions will promote student confidence and understanding.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students make connections between inequalities and real-world scenarios in 14.1.4 Try It and 14.1.5 Your Turn!. Students represent real-world contexts using inequalities and determine reasonable solutions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides an opportunity to engage students by editing questions to appeal to students’ interests. Consider informal observation to learn about students’ interests. For example, in question 1 in 14.1.4 Try It, change the item from a bike to an item most of the class wants.

Consider providing pictures or videos of the context in 14.1.4 Try It to provide a visual entry point.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

14.1.1 Warm-Up: Jumbotron Estimation

Component Overview: *Students estimate how many TV screens are needed to cover the jumbotron at Doak Campbell Stadium in Tallahassee.*

14.1.1 Warm-Up: Jumbotron Estimation

In the U.S., the average LCD TV screen has a width of 3.63 feet (ft) and a height of 2.04 ft.

Doak Campbell stadium has a jumbotron screen with a width of 119.7 ft and a height of 56.7 ft.

1. Estimate how many TV screens can be laid on top of the jumbotron at Doak Campbell stadium.
   Answers will vary. $\frac{(119.7)(56.7)}{(3.63)(2.04)} \sim 916$

   $119.7 \div 3.63 \sim 32$ *or* 33 *or* $56.7 \div 2.04 \sim 27$ *or* 28.

   $(32 \cdot 27) = 864$, *or* $(33 \cdot 28 = 924)$

   You can fit 864 *tv's to* 924 *tv's.*

2. Explain how you found your estimate.
   I found the area of the jumbotron and divided it by the area of the LCD TV.

Consider sharing pictures or videos of a jumbotron and a typical LCD TV. This will provide a visual entry point and help students assess the reasonableness of their estimation.

Doak Campbell Stadium is the home field stadium for the college football team at Florida State University. Show an image or video for any student fans in your classroom.

14.1.2 Exploration: Representing Inequalities on a Number Line

Component Overview: *Students explore the similarities and differences of solutions to inequalities and strict inequalities. Students analyze a visual representation of how to represent inequalities and strict inequalities on a number line, then practice graphing an inequality and strict inequality on a number line. Students then connect inequalities to real-world scenarios in which they reason about potential solutions and whether they make sense given the real-world context presented.*

14.1.2 Exploration: Representing Inequalities on a Number Line

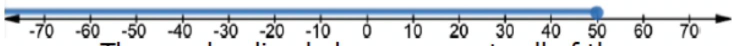
1. Complete the following problems.
   a. Choose an x-value that will make the inequality $x \leq 50$ true. *Answers will vary.* $x = 25$ *or* $x = 37$

   b. Choose an x-value that will make the inequality $x \leq 50$ false. *Answers will vary.* $x = 58$ *or* $x = 103$

   c. Have your partner verify your choices are correct.

   d. What is the x-value that is true for $x \leq 50$ but false for $x < 50$? 50

   e. The number line below represents all of the x-values that are true for $x \leq 50$.



   The number line below represents all of the x-values that are true for $x < 50$.

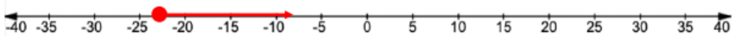
   How is the graph for $x < 50$ different from the graph for $x \leq 50$? *$x < 50$ does not include 50.*

2. Complete the following.
   a. How are the solutions to $x \geq -23$ and $x > -23$ the same?
   All values greater than −23 will be solutions.

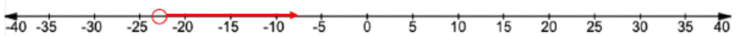
   b. How are the solutions to $x \geq -23$ and $x > -23$ different?
   $x > 23$ does not include −23.

   c. Graph the solutions to each inequality on the number lines.

$x \geq -23$



$x > -23$



Think-Pair-Share
Have students first complete the questions independently and then discuss their work with a partner. These discussions will promote student confidence and understanding.

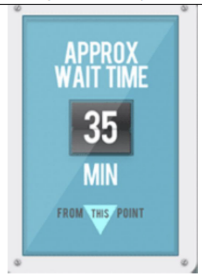
Please read the following inequality: $x \geq 23$
- *What does the line under the inequality symbol mean?*
- *How do you represent this inequality on the number line?*

Please read the following inequality: $x > 23$
- *How do you represent this inequality on the number line?*
- *What is the difference between these two inequalities?*
- *How does the difference impact the solutions to these inequalities?*

14.1.2 Exploration: Representing Inequalities on a Number Line (continued)

3. Complete the following.

   a. With your partner, determine which inequality best represents each context.

Nutritional Information Calories: 2,000 Total Fat Less than 65g Sat Fat Less than 20g Cholesterol Less than 300mg Sodium Less than 2,400mg **Total Fat** $f \le 65$ $f \ge 65$ $f < 65$ $f > 65$	YOU MUST BE AT LEAST **54"** TALL TO RIDE $h \le 54$ $h \ge 54$ $h < 54$ $h > 54$
 $p \le 10$ $p \ge 10$ $p < 10$ $p > 10$	 $s \le 50$ $s \ge 50$ $s < 50$ $s > 50$
 $x \le 50$ $x \ge 50$ $x < 50$ $x > 50$	 $w \le 35$ $w \ge 35$ $w < 35$ $w > 35$ *Answers will vary*

Consider opportunities for engaging students with real-world examples of this activity. Does your school or classroom have a collection of signs like these around? Conduct a scavenger hunt. Could students bring in their own nutritional food labels? Conduct a student-led activity on what they bring to class instead.

14.1.2 Exploration: Representing Inequalities on a Number Line (continued)

b. In which, if any, of these situations does a value of 0 make sense given the context?
   Total fat

c. In which, if any, of these situations does a negative number make sense given the context?
   None

d. In which, if any, of these situations does 1,000 make sense given the context?
   None

Have students write similar questions like questions 3a-3d for the other signs they may have collected earlier in 14.1.2 Exploration's discussions. Then have students trade their questions with a partner.

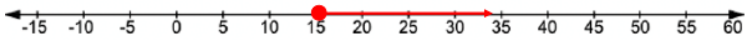
14.1.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

Component Overview: *Students represent real-world scenarios using inequalities and number lines. Then students determine possible solutions to the inequalities. Remind students to consider the context when determining possible solutions.*

14.1.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

1. According to the Interstate Highway Standards, U.S. and state highway traffic lanes must be at least 15 feet wide.
   a. Write an inequality to represent the width, w, that traffic lanes can be. $w \geq 15$

   b. Represent the inequality on the number line.

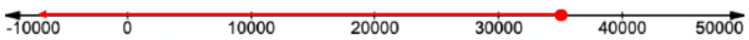


   c. List two examples of possible widths, in feet, for the traffic lane. *Answers will vary.* $w = 15.25$ *ft or* $w = 15.3$ *ft*

   d. Give one example of a width that is true for the inequality but does not make sense in the real world. 18 *ft*

2. High-altitude parachute jumps are usually made for an altitude that is no more than 35,000 feet to avoid frostbite from extreme cold.
   a. If h represents the height of a jumper, write an inequality to describe all possible values of h. $h \leq 35{,}000$

   b. Represent the inequality on the number line.



   c. Does the inequality that represents the situation include 0 as a possible height? *Yes*

   d. Explain why a negative number does not make sense for this context. *One would not jump with a parachute from a negative height since this is below sea level.*

Read each scenario aloud. Then lead a discussion focused on preventing potential misconceptions.

- *What does it mean if something is at least 15 feet wide? Is it larger or smaller than 15? What are three possible solutions?*
- *How do you represent the inequality on a number line?*
- *What does it mean if the jumps are no more than 35,000 feet?*
- *What is the greatest distance possible? What are three possible solutions? How do you represent this on the number line?*

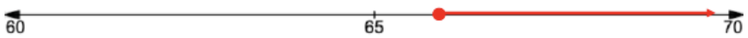
14.1.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

Component Overview: *Students independently represent real-world scenarios using inequalities and number lines. Then students determine possible solutions to the inequalities based on the context.*

14.1.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

1. The inequality $67.85 \leq m$ represents the amount of money, m, Edin needs to earn to buy a new bike.

   a. Graph the inequality.



   b. Give one value that is true for $67.85 \leq m$ but is unreasonable for the context.
   $\$700$

   c. Explain why the value you chose is unreasonable.
   Edin would have enough money for several bikes.

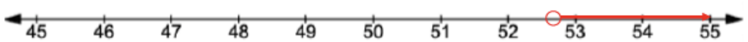
2. A farmer plans to plant more than 52.75 acres of oranges.

   a. If A represents the number of acres of oranges the farmer plants, write an inequality that represents the number of acres the farmer plants.
   $A > 52.75$

   b. Circle all the possible values for A, the number of acres the farmer plans to plant, from the list below.

52.7 acres	$52\frac{3}{4}$ acres	53 acres
52 acres	52.8 acres	$52\frac{1}{4}$ acres

   c. Graph the inequality.



Consider editing questions or allowing students to adjust contexts to appeal to students' interests. For example, students could adjust question 2 to be "You need to have more than $52.75 to buy ________."

If students struggle to determine unreasonable values, pose scaffolding questions, such as:

- *What is the minimum amount that Edin needs?*
- *How much greater or less than the minimum amount would be a reasonable answer?*

Challenge students to also determine unreasonable values for the acres too. Ask students to explain why these values are unreasonable based on the context.

14.1.5 Your Turn!

Component Overview: *Students work independently to determine if they would rather ride a go-cart or an electric bike based on the speed of each. Students determine possible speeds and plot the given inequalities on number lines.*

14.1.5 Your Turn!

1. Would you rather drive a go-cart whose speed, g, in miles per hour (mph), is represented by the inequality $12.5 \geq g$, or an electric bike whose speed, b, in mph, is represented by $b \geq 7\frac{1}{2}$?

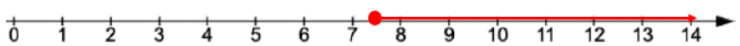
   a. I choose the ○ go-cart ○ electric bike because ________ *Answers will vary*.

   The ○ go-cart ○ electric bike could have a speed of 18 mph.

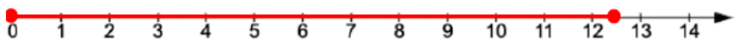
   b. Circle all the values from the list below that are possible speeds for the go-cart.

1.25 mph	15.2 mph	$10\frac{1}{3}$ mph	$\frac{12}{5}$ mph
12.45 mph	11.95 mph	$12\frac{2}{3}$ mph	$11\frac{11}{12}$ mph

   c. Represent the speed of the electric bike on the number line.



   d. Represent the speed of the go-cart on the number line.



   e. Explain why the number lines do not include negative numbers.
   You cannot go negative speeds.

Students may struggle to represent the speed of the go-cart on the number line because they have not represented a compound inequality yet. Discuss what negative numbers mean in this context and how they could be represented on the number line.

14.1.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate understanding of writing an inequality, determining possible values that satisfy the inequality, and representing the inequality on a number line.*

14.1.6 Wrap-Up: Ticket Out the Door

At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

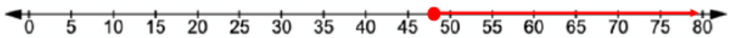
1. Write an inequality representing the height, h, a person must be to go on the ride.

   $h \geq 48$

2. Circle all values from the list below that satisfy this inequality.

47.9 in.	100 in.	$\frac{48}{2}$ in.	48.5 in.
$70\frac{3}{4}$ in.	$48\frac{1}{4}$ in.	0 in.	400 in.
48.099 in.	48 in.	$50\frac{3}{4}$ in.	47.9 in.

3. Graph the inequality on the number line.



Consider challenging learners to write an explanation of how they determined their answers (i.e. I tested each value and read the inequality out loud to see if it was true).

14.2 Finding an Unknown Value in an Equation

Lesson Introduction

Benchmark	MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.	Learning Targets	• I can determine an unknown decimal in a numeric equation with three numbers. • I can determine an unknown fraction in a numeric equation with three numbers.

Benchmark Coherence	Building On	Working Toward
	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	N/A

Essential Questions	• How can you represent unknown values using variables? • How can you use reasoning to determine the value of an unknown number in an equation? • How can you use a bar model to determine the value of an unknown number in an equation? • How can you use a diagram to determine the value of an unknown number in an equation? • Why do different equations have the same solution?
Lesson Narrative	This lesson builds on students' understanding to solve one-step equations with fractions and decimals using any of the four operations. The equations relate three numbers, with the unknown on either side of the equal sign in any position. The lesson emphasizes using reasoning, diagrams, and mental math to figure out the unknown instead of using inverse operations. All terms and solutions will be positive, rational numbers.

Component Summary				
	14.2.1 Warm-Up (~5 min.)	Students reflect on different careers in STEM *(SE p. 342)*	14.2.4 Collaboration (10 min.)	Students discuss how to solve equations using reasoning and diagrams *(SE p. 345)*
	14.2.2 Refresher (5 min.)	Review identifying variables and algebraic equations *(SE p. 342)*	14.2.5 Your Turn! (10 min.)	Practice figuring out the unknown in equations *(SE p. 346)*
	14.2.3 Guided Instruction (15-20 min.)	Solve equations using reasoning and diagrams *(SE p. 343)*	14.2.6 Wrap-Up (~5 min.)	Students reflect on what they learned and what they still have questions about *(SE p. 347)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: • Variable • Equation • Algebraic equation • Solve • Solution <u>Additional Terms:</u> N/A	• Career profile video • (Optional) 10×10 grids • (Optional) Fraction tiles	• 14.2.1 Warm-Up contains a video to accompany the question prompts.

Teacher Tips

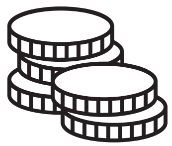
Common Misconceptions	Things to Consider
• Students may think they should use the operation and numbers presented in the equation rather than thinking about finding the value of the unknown. • Students may perform the calculations incorrectly if using mental math when the values do not have the same number of decimal places. • Students may believe coefficients are added to the variable rather than the multiplicative relationship of coefficient and variable. • Students may add or subtract across numerators and denominators rather than finding a common denominator or thinking about the values represented.	• Encourage students to draw their own diagrams to model the problems if they have trouble identifying with or understanding the diagrams in the lesson. • Encourage students to see if their solutions make sense in the original problem before substituting to check. Then encourage students to substitute their solutions back into the original equation to check. • Have templates of 10×10 grids available if students need to use them. • This lesson is an opportunity to further build students fluency with the operations of rational numbers. Students should not have access to calculators for this lesson. • Students should not be formally using properties of equality to solve the equations. The lesson is meant to encourage algebraic reasoning through models. • If time is an issue, then assign 14.2.4 Collaboration and/or 14.2.5 Your Turn! for homework.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Algebra Talk Consider utilizing Algebra Talk with the problems in 14.2.3 Guided Instruction. After presenting a problem, give students time to formulate strategies to figuring out the unknown in the equation. Ask different students how they would approach a problem. Then discuss the strategies as a group and decide how to proceed.	**MA.K12.MTR.2.1: Multiple Representations** Students will be working to figure out an unknown variable in multiple ways. Clarifications within this standard emphasize building understanding through modeling. Utilize the progression of the lesson to highlight the process by which students are moving from equation to diagram to solution.

Differentiate Instruction

Consider the following strategies for additional entry points for all learners:

- If play money is available, struggling students might find it helpful to use the money to help solve money-related problems. Additionally, coins like dimes, nickels, quarters, and pennies might help students make better sense of decimals. Students may not initially make sense of tenths, but they can articulate the worth of a dime compared to a dollar.
- 10×10 grids may be helpful for struggling students. Have blank 10×10 grids available for use.



Lesson Notes:

14.2.1 Warm-Up: What Is STEM?

Component Overview: *Play the Career Profile video on STEM for the entire class. After students watch the video (3:06 minutes), prompt them to reflect on the profession by answering the two questions independently.*

14.2.1 Warm-Up: What is STEM?

Science, Technology, Engineering, and Mathematics (STEM) is a way of thinking and doing. STEM skills are useful to almost every career.

An individual does not need to be a master in all four areas of STEM to be successful; teamwork plays an important role. Playing to a person's strengths and working as a team creates a greater impact than working alone.

1. Describe a time where it was better for you to work as a team than alone.

   Answers will vary.

2. List two jobs that involve working as a team but are in different fields of STEM.

   Answers will vary. Engineers and software designers both work in teams on projects.

Consider strategies for highlighting STEM in your classroom and inspiring students for STEM work.

- *Do you know someone who works in a STEM field? Have them visit or virtually meet to answer questions about their job.*
- *Reach out to a local STEM company and ask if they do school or community work. Most companies would love to partner with a classroom.*
- *Consider having students return to this lesson later in the year for project-based learning around the importance of teamwork in the workplace.*

14.2.2 Refresher: The Unknown Value

Component Overview: *Students will review variables and their role in algebraic expressions and equations. Remind students that any letter can be used in any situation.*

14.2.2 Refresher: The Unknown Value

With your partner, complete the following.

In math, **variables** are used to represent unknown values.

In an algebraic expression, a ***variable*** is a letter that represents a number.

1. Circle the variable(s) in each expression.

$m + 5$ $\qquad$ $2x + y$ $\qquad$ $\frac{a}{4}$

Variables are found in algebraic equations.

An ***equation*** is a mathematical relation statement where two equivalent expressions or values are separated by an equal sign.

An ***algebraic equation*** is an equation with at least one variable.

2. Circle the algebraic equations.

$6 + 8 = 14$ $\qquad$ $5y = 35$ $\qquad$ $2w + 7 = 13$ $\qquad$ $\frac{12}{3} + 8 = 12$

To **solve** an equation means to find the solution.

The **solution(s)** to an algebraic equation is(are) the value(s) of the variable(s) that makes the equation true.

Consider reviewing variables, equations, and expressions with students to activate prior knowledge. Remind students that an equation contains an equal sign, whereas an expression does not have an equal sign. Write examples of equations and expressions on the board and ask students to sort them accordingly.

Prompt students to activate prior knowledge of the definition of equation, which was taught in the previous unit. This is the first time students will formalize the definitions for variable and algebraic equations. After defining as a class, provide examples and nonexamples for students to consider. Position students to deepen understanding and address misconceptions through questions like:

- *What is the role of a variable in an expression or an equation?*
- *Could any letter be used as the variable?*
- *How is an empty box similar to a variable?*

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

Component Overview: *Students will begin exploring how to use reasoning and visual aids to solve algebraic equations. Tell students that they can use multiple methods to solve an equation, including logic, bar diagrams, grids, and other visual aids. Remind students that they are solving for the unknown number to make an equation true.*

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

For some equations, the solution can be found mentally or by using reasoning.

1. The solution to the equation $n - 12.1 = 35.3$ can be found using reasoning.
   a. iscuss with your partner how to find the value of n in the equation $n - 12.1 = 35.3$.
   Answers will vary. Add 12.1 and 35.3. Substitute values for n until you find one that makes the equation true.
   b. The equation is rewritten vertically with a box in the place of the variable. Write the solution in the box and check whether it makes the equation true.

$$\begin{array}{r} \boxed{47.4} \\ \underline{-\ 12.1} \\ 35.3 \end{array}$$

Diagrams can also be helpful to find a solution.

2. Olivia gave equal amounts of money to each of her 3 children. Olivia gave each child $\$16.75$.
   The bar model represents the equation $\frac{m}{3} = 16.75$, where m is the total amount of money Olivia gave all of her children.

16.75	16.75	16.75
m		

   a. How can the bar model be used to find the total amount of money Olivia gave all of her children, m?
   Add $16.75 + 16.75 + 16.75$ or multiply 16.75×3 to find m, the total amount of money.
   b. What number can be divided by 3 to get a result of 16.75?

$$\frac{\boxed{50.25}}{3} = 16.75$$

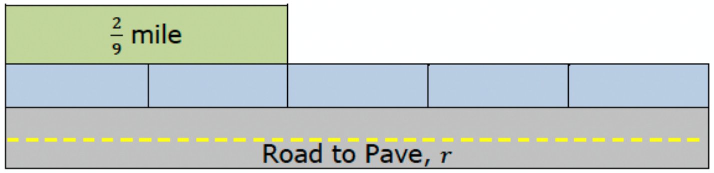
Consider allowing students to create a diagram of the equation on a 10×10 grid. Encourage students to explain how their diagram relates to the equation.

Use correct vocabulary when reading the decimal numbers and hold students responsible for doing so as well. If needed, review decimal terminology and tenths with students.

Consider using the following prompts to encourage students to reason through the problem:
- *How do you think you could determine the value of the variable?*
- *Why is it helpful to substitute the value back into the equation?*

Algebra Talk
Consider using Algebra Talk as a way to encourage students to approach problem-solving in a variety of ways. Ask different students how they think the problem could be solved. Then, as a class, discuss the options presented and decide together how to proceed.

3. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9}=\frac{2}{5}r$, where r is the length of the whole road being paved.



a. Discuss with your partner how the diagram represents the information given.

Answers will vary. The bottom and middle bars represent the entire amount to be paved. The top bar shows the amount that has already been paved.

b. Describe how to use the bar model to solve the equation for r, the length of the road being paved.

Answers will vary. Use the top bar to figure out the length of the unpaved section and then add the amounts together.

c. How long is the road being paved?

$\frac{5}{9}$ mile

d. Check your solution. $\frac{2}{9}=\frac{2}{5}\left(\boxed{\frac{5}{9}}\right)$

Encourage students to verbally explain the problem. Have students tell a partner what information is given and what information they need to find. Discuss how to use the bar model and if another approach may also work to solve the problem.

Scaffold instruction through intentional questioning.
$\frac{2}{9}$ mile is $\frac{2}{5}$ of the entire paving job.

- *Why do we use multiplication, $\frac{2}{5}r$, in the equation?*
- *How can we use the drawing to help us find the solution?*
- *Does your answer make sense?*
- *How can you check your solution?*

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams (continued)

4. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.

The equation $m \div 3 = \frac{3}{4}$ can be used to find m, the amount of milk Donté used.

a. Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.

$$\frac{m}{3} = \frac{3}{4}$$

$$m = (3)\frac{3}{4} = \frac{9}{4} = 2\frac{1}{4}$$

b. Check your solution. $\boxed{2\frac{1}{4}} \div 3 = \frac{3}{4}$

For question 4a, students may solve by adding ¾ + ¾ + ¾ and that is fine. They do not need to use the inverse operation as they haven't learned this yet.

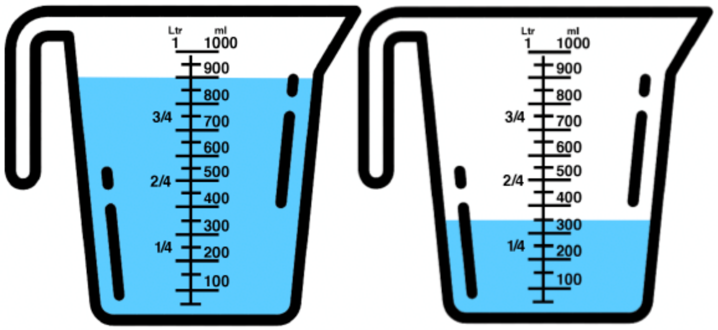
14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

Component Overview: *In the activity, students will work with a partner to solve algebraic equations by using diagrams such as measurement and bar models. Problems will use decimal numbers and fractions. This activity is a collaborative exercise, so students can either work in pairs on the same problem or work in pairs on different problems and then compare answers.*

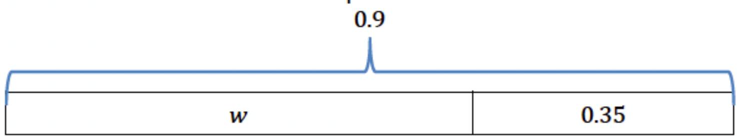
14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

With your partner, solve the following problems using reasoning and diagrams.

1. Vincent had 0.9 liters (L) of water. He poured some into the mixing bowl and had 0.35 L left.



The bar model shown represents this situation.



a. Discuss with your partner how the bar model represents Vincent's situation. Then, write a brief summary of your discussion.

The entire 0.9 bar represents the total amount of water Vince had. The 0.35 portion shows the amount he poured into the mixing bowl. The bar labeled w represents the amount of water Vince has left.

Notice and Wonder

- *What do you notice about the representations of the problem?*
- *What is the same? What is different?*

If needed, review liters with students and show students an actual liter for assistance in sensemaking.

14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams (continued)

b. The equation $0.9 - w = 0.35$ can also be used to represent this situation. Find the value of w, the amount of water that Vincent poured into the mixing bowl.

$w = 0.55$ *L*

2. Josephine told her brother she would give him $\frac{1}{2}$ of the money they made from hosting a lemonade stand at their school carnival. She gave her brother $\$24.25$. This situation can be represented by the equation $\frac{1}{2}x = 24.25$ to find x, the amount of money they made at the lemonade stand.

a. Use the bar model shown to represent the situation.

$\$24.25$	$\$24.25$
x	

b. Find the value of x, the amount of money they made at the lemonade stand.

$\$48.50$

Have students make connections between the equation and the diagrams above. Encourage students to explain what each part of the equation represents and how it relates to the diagrams.

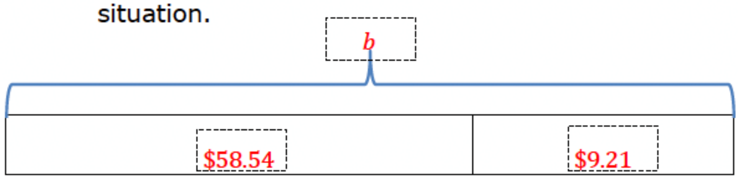
14.2.5 Your Turn!

Component Overview: *Students will practice solving equations with reasoning and diagrams. Each problem requires students to work with different operations. Students will work with decimal numbers when solving the problems. Remind students that they should include units in their answers, whether the units are monetary or measurement.*

14.2.5 Your Turn!

1. DeAndré paid $58.54 for a game at Store A, which was $9.21 less than the price of the game at Store B. This situation can be represented by the equation $58.54 = b - 9.21$, where b is the price of the game at Store B.

    a. Complete the bar model diagram to represent this situation.

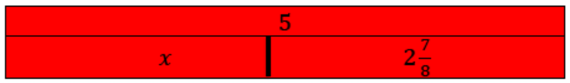


    b. Find the value of b.

    $67.75

2. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x, the gallons of water dispensed from the bottle.

    a. Draw a diagram representing the situation.



    b. How much water was dispensed from the bottle? Include units.

    $2\frac{1}{8}$ *gallons*

- *How does the bar diagram help you think about how to find the solution to the given equation?*
- *Why do you think the "58.54" part of the bar is longer than the "9.21" part?*
- *Why is the answer for question 1b the sum of 58.54 and 9.21?*

Students may need a quick review or explanation of what the word "dispensed" means to clear up any contextual stumbling blocks to finding the answer.

14.2.6 Wrap-Up: Reflection

Component Overview: *This Wrap-Up encourages students to assess their own learning. Students will think about additional questions they have, what they have learned, and what they are still unsure of regarding solving equations by using reasoning and diagrams. Remind students to honestly answer the questions as a reflection of their learning.*

14.2.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered…

Answers will vary but should apply to the lesson.

One mistake I learned from today…	One thing I am not sure about…YET!
Answers will vary but should apply to the lesson.	*Answers will vary but should apply to the lesson.*

One thing I am confident I know is…

Answers will vary but should apply to the lesson.

For students who might struggle in representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson.

- *Leverage sentence stems like the Learning Targets or Essential Questions for students to reference to provide more precise responses.*
- *Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example.*
- *Mark-Up Text: Students can mark each activity with symbols that represent each of the boxes provided and explain why they chose that symbol for that activity in the corresponding box.*

14.3 Discovering Inverse Operations

Lesson Introduction			
Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.	Learning Targets	• I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations. • I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.

Benchmark Coherence	Building On	Working Toward
	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• What are inverse operations? • How would you use inverse operations to solve an equation? • How would models help you solve an algebraic equation using inverse operations?
Lesson Narrative	In this lesson, students apply and extend previous understandings of arithmetic to algebraic expressions. Using visual models like algebra tiles and bar models and scales to represent abstract concepts of operations and equations, students explore using addition and subtraction as inverse operations to solve one-step equations and use manipulatives to discover the role of inverse operations to solve one-step equations.

Component Summary				
	14.3.1 Warm-Up (~5 min.)	Create a visual representation to solve a real-world problem *(SE p. 348)*	14.3.4 Guided Instruction (15-20 min.)	Modeling solving equations with inverse operations *(SE p. 350)*
	14.3.2 Exploration (10 min.)	Solve equations using algebra tiles *(SE p. 348)*	14.3.5 Your Turn! (10-15 min.)	Solve problems using algebra tiles, diagrams, and inverse operations *(SE p. 352)*
	14.3.3 Guided Instruction (5 min.)	Solve equations using inverse operations *(SE p. 350)*	14.3.6 Wrap-Up (~5 min.)	Self-reflect on the learning of the lesson *(SE p. 354)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Inverse operation • Addition Property of Equality • Subtraction Property of Equality Additional Terms: • Opposite	• (Optional) Algebra tiles • (Optional) Blank tape diagrams • (Optional) Balance scale • (Optional) Marbles	N/A

Teacher Tips

Common Misconceptions

- Even though “opposites” and “inverse” can be used interchangeably in this lesson, be cautious that students will associate “inverse operations” with “opposites” in all cases.
- Students may not have much experience with variables and be confused by a letter standing in for a number.
- Students may struggle with integer operations.
- Students may struggle to understand that what happens on one side of an equation must also occur on the other side to keep the equation balanced.

Things to Consider

- Emphasize the visual of a balance beam or a hanger to help students understand and remember that meaning of the equals sign and that an operation that is done to one side of the equals sign of an equation must also be done on the other side of the equal sign.
- Consistently switch the order of “addition” and “subtraction” in discussion to instill that both operations are important and relate to one another.
- Tell students that while “opposites” and “inverse” can be used interchangeably in this lesson, that is not always the case. In mathematics, “opposites” refer specifically to two numbers whose sum is zero.
- Highlight opportunities to clarify the distinction between “inverse operations” and “opposite values.” For example, in question 3 in 14.3.4 Guided Instruction, for $x + 5 = 11$, the “$+5$” represents both the inverse and opposite.
- If time is a concern, consider assigning 14.3.5 Your Turn! for homework.

Literacy and Language Routines

Notice and Wonder

14.3.2 Exploration positions students to use algebra tiles to represent and solve algebraic equations. Consider applying Notice and Wonder routines on each set of tiles, positioning students to discover the value of the variable through observation of the algebra tiles.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.6.1: Reasonableness

Students will be working to use inverse operations to solve one-step equations involving addition or subtraction. Clarifications within this standard emphasize the appropriateness of a solution to a problem. Encourage students to substitute their solution back into a problem and to verify their solution by explaining the method used.

Differentiate Instruction

Explain that the bar models, balance, hanger, and number line are all different ways of modeling the same concept of balancing an equation. Consider allowing students to use whichever method makes the most sense to them, like an actual hanger or balance to model the concept of balancing an equation by adding to or removing objects from each side of it. Allow students to use visual aids to solve equations for as long as they need to understand the concept. Encourage students to explain the process they use to solve the equations.

Lesson Notes:

14.3.1 Warm-Up: Picture This

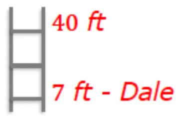
Component Overview: *Students draw a visual representation of a mathematical situation and then use the image to solve the problem. Encourage students to explain the reasoning they used to solve the problem.*

14.3.1 Warm-Up: Picture This

Dale is helping to fix the roof at his uncle's house. They are using a 40-foot ladder. Dale is climbing the ladder to reach the roof. He is currently 7 feet up the ladder.

1. Sketch a picture to represent this situation.

   Answers will vary.



2. How much farther does Dale have to go to reach the roof? Include units.

   33 feet

After students complete the Warm-Up, have them explain how they arrived at their answer. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as:

- *"I drew…"*
- *"I figured out…"*

14.3.2 Exploration: Solving Equations Using Algebra Tiles

Component Overview: *Students use algebra tiles to understand the process of using inverse operations to solve algebraic equations. Students compare the number of algebra tiles on each side of an equation to determine the value of the variable tile. Students then discuss the equivalent relationship with a partner.*

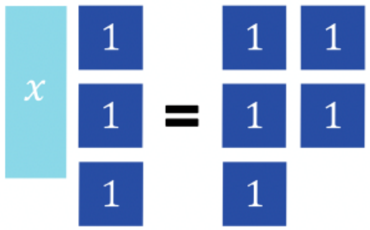
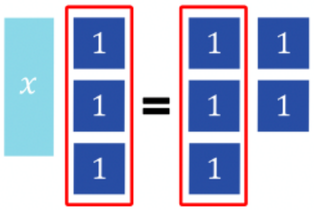
14.3.2 Exploration: Solving Equations Using Algebra Tiles

Algebra tiles can be used to represent and solve algebraic equations.

In the model shown, 1 represents 1 and x represents x.

1. The steps to determine the value of x using algebra tiles are shown.

   a. Work with your partner to complete the last step according to the description.

Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

Consider allowing students to use algebra tiles to recreate the problem. Prompt students to discuss their rationale and approach as they work to continue building mathematical literacy.

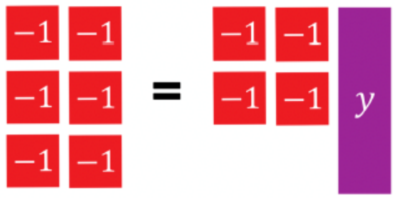
14.3.2 Exploration: Solving Equations Using Algebra Tiles (continued)

b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.
Answers will vary. The tiles I drew represent an equivalent relationship because the original problem $x + 3 = 5$ represented an equivalent relationship, and since I removed the same number of tiles from each side of the equation, the relationship will still be equivalent.

c. How many blue tiles are equivalent to one teal tile?
2 blue tiles are equivalent to one teal tile.

d. What is the value of x in the equation $x + 3 = 5$?
$x = 2$

2. In the model shown, $\boxed{-1}$ represents -1 and $\boxed{y}$ represents y.



a. Work with your partner to determine how many red tiles are equivalent to one purple tile.
2 red tiles are equivalent to one purple tile.

b. What is the value of y in the equation $-6 = -4 + y$?
$-2 = y$

Consider giving students a blank tape diagram. Have students use the diagram to build a model of the equation. Encourage students to explain how and why they are using the pictured algebra tiles to create a bar diagram.

Prompt students to make connections between their visual models and the learning targets for this lesson.

- *What do you notice about the equation?*
- *How does the equation relate to the tiles?*

14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

Component Overview: *Students are introduced to the concept of using inverse operations to solve one-step equations. Students begin exploring the inverse operations of addition and subtraction.*

14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

   Answers will vary. For an equation to remain balanced, you must add or subtract exactly the same quantity from both the left and the right side of the equation.

When finding the solution to an equation, inverse operations are used to balance an equation.

An ***inverse operation*** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction $(-)$	*Addition* $(+)$
Addition $(+)$	*Subtraction* $(-)$

Work with students to define inverse operations. Discuss how the word "inverse" can be used in other contexts. Review the relationship of addition to subtraction to help students understand why inverse operations work when solving an equation.

Consider showing students inverse operations using arithmetic operations. Empathize the role of the equal sign.

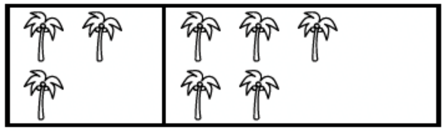
14.3.4 Guided Instruction: Solving Equations With Inverse Operations

Component Overview: *Students will begin exploring how to use inverse operations to solve one-step equations involving addition or subtraction. Students will use bar models and a balance scale to explore balancing on each side of an equation.*

14.3.4 Guided Instruction: Solving Equations with Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.



The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$3 + \underline{\ 5\ } = 8$

$8 - 3 = \underline{\ 5\ }$

After students create the equations, have them discuss how the equations relate to each other. Guide students toward the idea of addition and subtraction being inverse operations and what that means.

2. The bar model shown can be used to represent either an addition or subtraction equation.

7	
x	4

a. Circle the addition equation that matches the diagram.

$x + 7 = 4$ $\quad$ $7 = x + 4$ $\quad$ $x = 7 + 4$

b. Write an equivalent equation using the inverse operation.

Answers will vary. $x = 7 - 4$; $4 = 7 - x$

c. Justify to your partner that the equation you wrote is an equivalent equation.

Answers will vary. Since I used inverse operations to rewrite the equation, my equation is an equivalent equation because the value of the expressions on both sides of the equation remain equivalent.

To provide additional scaffolding for students who may be struggling to make connections and identify the corresponding equations, ask questions like:

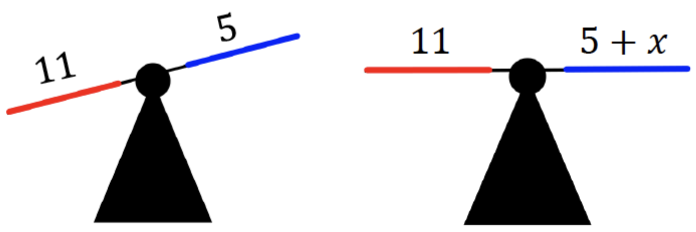
- *What do you notice about the bar model? What are the parts, and what represents the whole?*
- *How does the equation represent the bar model?*

Students could write $7 - 4 = x$ *OR* $7 - x = 4$*. Both would be correct, but one makes finding the value of* x *easier. Be ready to make those connections and listen for student thinking.*

14.3.4 Guided Instruction: Solving Equations With Inverse Operations (continued)

d. What is the value of the variable, x?
$x = 3$

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value, x, needs to be added to the right side of the scale.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of x by completing the steps shown.

$x + 5 = 11$

$x + 5 - \boxed{5} = 11 - \boxed{5}$

$x = \boxed{6}$

b. Rewrite the original equation, substituting the solution for x.

$5 + 6 = 11$

c. Explain how to use the equation in Part B to confirm that your solution is correct.
Answers will vary. The equation in Part B confirms that my solution is correct because the value of the expressions on both sides of the equal sign remain equivalent with my solution.

As the class works through question 3, use the following prompts to guide students toward making the connection between the visual of the balanced scales and balancing equations. This will build conceptual understanding that is foundational for lessons moving forward.

- *What do you notice about the balance scales? What is the same? What is different?*
- *How is the second balance scale related to the equation in question 3a?*

14.3.4 Guided Instruction: Solving Equations With Inverse Operations (continued)

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.

Addition Property of Equality

If $a = b$, then $a + c = b + c$

Subtraction Property of Equality

If $a = b$, then $a - c = b - c$

4. The equation $x + 23 = 78$ can be solved by using inverse operations.

   a. Fill in values in each equivalent equation to solve for x.

   $x + 23 = 78$
   $x + 23 - 23 = 78 - 23$
   $x = 55$

   b. What is the inverse operation used to solve the equation $x + 23 = 78$?
   Subtraction

   c. With your partner, discuss how to confirm whether or not the solution is accurate.
   Answers will vary. To confirm that the solution is accurate, I can substitute my solution into the original equation and confirm that the values on each side of the equal sign are equivalent to each other.

Refer to MA.K12.MTR.6.1 as you remind students to assess whether their solution is reasonable and to substitute their answer back into the equation as a way to verify their solution.

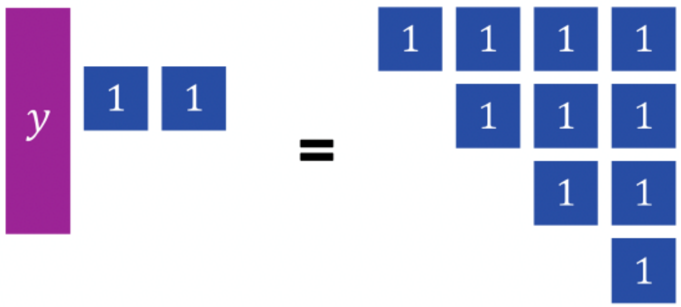
Have students construct an equivalent equation for $x + 23 = 78$ and explain how and why the equations are equivalent. Also encourage students to construct another pair of equivalent equations using addition or subtraction and then explain how inverse operations can be used to solve them.

14.3.5 Your Turn!

Component Overview: *Students will practice using the inverse operations of addition and subtraction to solve one-step equations.*

14.3.5 Your Turn!

1. The algebra tiles below model an equation.



   a. How many blue tiles are equivalent to one purple tile?
   8 *blue tiles are equivalent to one purple tile.*

   b. Write a one-variable equation to represent the tiles on the mat.
   $y + 2 = 10$

   c. What inverse operation can be used to solve the equation?
   Subtraction

   d. What value of y makes the equation true?
   $y = 8$

2. The equation $-31 = s - 6$ can be solved using inverse operations.
   a. Fill in the missing information to solve for s.

$$-31 = s - 6$$
$$-31 + 6 = s - 6 + 6$$
$$-25 = s$$

Consider an inquiry-based learning approach to facilitate connections between visual models and abstract concepts.

- *How can you figure out the relationship between the blue tiles and purple tiles?*
- *How can you use this relationship to solve the equation?*
- *Why do you use inverse operations to solve the equation?*

b. Which inverse operation was used to solve the equation?
Addition

c. Check your solution.
$-31 = -25 - 6$
$-31 = -(25 + 6)$
$-31 = -31$

3. Two bar model diagrams are shown.

Bar Model Diagram 1	Bar Model Diagram 2
m / -12 | -4	-12 / 4 | m

a. Write an algebraic equation for each diagram box diagram.

Algebraic Equation for Diagram 1	Algebraic Equation for Diagram 2
Answers will vary. $-12 + (-4) = m$ $m = -4 + (-12)$	*Answers will vary.* $4 + m = -12$ $-12 = m + 4$

b. Find the value of variable, m, for each algebraic equation.

Diagram 1	Diagram 2
$-16 = m$	$m = -16$

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.
Answers will vary. I noticed that that the solutions to both equations are the same even though the equations have different values.

Solving this question requires students to apply the inverse operation of addition to both sides of the equation. Reiterate the importance of assessing the reasonableness of a solution and substituting the solution back into the equation as a way to check it.

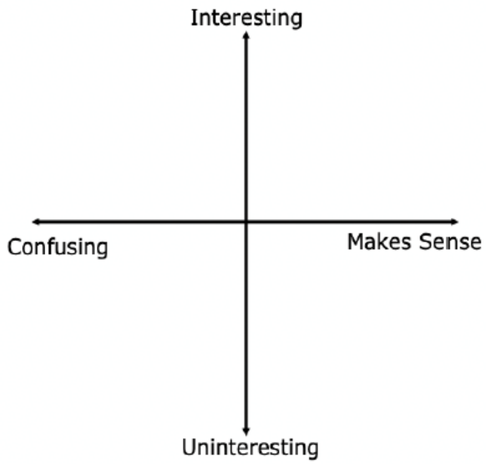
14.3.6 Wrap-Up: Reflection

Component Overview: *Students will reflect on their learning and understanding of the lesson. Encourage students to not only plot a point representing how they feel about meeting the learning target but to also explain why they feel that way.*

14.3.6 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."

   Answers will vary.



Consider prompting students to include the learning target as an ordered pair and explaining why they chose that value. For example, a student may choose $(-1, 5)$ instead of $(-4, 5)$, explaining that it was confusing at first but made more sense by the end of the lesson.

14.4 Solving One-Step Equations in One Variable Using Addition and Subtraction

Lesson Introduction

Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.	Learning Target	• I can solve addition and subtraction one-step equations.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• How do you solve one-step addition or subtraction equations? • How can you use models to solve equations? • How do you know two equations are equivalent?
Lesson Narrative	Students will use inverse operations to begin solving algebraic equations without the support of models. The lesson begins with the support of visual models and then moves to solving addition and subtraction equations algebraically Students begin to explore equivalent equations to extend their algebraic reasoning.

Component Summary	14.4.1 Warm-Up (~5 min.)	Visualize solutions *(SE p. 355)*	14.4.4 Exploration (10 min.)	Solve equations using opposites *(SE p. 358)*
	14.4.2 Collaboration (10 min.)	Create algebraic equations from a diagram *(SE p. 355)*	14.4.5 Collaboration (5 min.)	Solve equations *(SE p. 359)*
	14.4.3 Guided Practice (10 min.)	Solve one-step algebraic equations using inverse operations *(SE p. 357)*	14.4.6 Your Turn! (5 min.)	Solve one-step equations using addition or subtraction *(SE p. 360)*
	14.4.7 Wrap-Up (~5 min.)	Write a note of how to use inverse operations to solve an equation *(SE p. 360)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms:</u> • Inverse operations • Opposites	• (Optional) Counters • (Optional) Algebra tiles	N/A

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may struggle with understanding equivalent equations such as $x - 4 = 3$ is equivalent to $-4 + y = 3$. • Students may struggle with integer operations.	• Draw a circle around the equal sign to emphasize that an equation is made of two equivalent expressions. • Use diagrams and models to illustrate equations and the process of solving them as much as needed. • As students continue to gain an understanding of solving equations students may struggle with solving equations with negative numbers. This struggle may be because they struggle with operations with integers or it may because they do not grasp the mathematical meaning of an inverse. Reviewing the additive inverse property may be helpful. Students may struggle when equations are written in different forms. For example, $-2 + y = 3$ in 14.4.4 Exploration question 8, students may subtract 2 from both sides.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display While pairs are working, circulate, collect, and make a visual display of vocabulary, phrases, and representations students use as they solve each situation. Make connections between how similar ideas might be communicated and represented in different ways.	**MA.K12.MTR.4.1: Discussion and Reflection** Students will be working to use inverse operations to solve one-step equations. Clarifications within this standard emphasize analyzing mathematical thinking and explaining methods and procedures. Encourage students to reflect on the processes they use and to describe them using correct mathematical vocabulary.
Differentiate Instruction	Operations with integers may still require a variety of entry points for students at different moments in their mathematics journey. For example, in the 14.4.1 Warm-Up, a student may know that the operation is addition and that the inverse is subtraction and therefore be able to answer all essential questions. However, when calculating, they may not get 25. Consider the strengths in this student and provide scaffolding for that student to still find success in this lesson. Whether using a calculator, accepting explanations to answers, or utilizing On-Ramp video learning tools and practice, consider additional entry points for learners at all stages in their journey.	
Lesson Notes:		

14.4.1 Warm-Up: Visualizing Solutions

Component Overview: *Students create a visual model to represent an equation. Students will then use the model and the equation to solve for the missing addend.*

14.4.1 Warm-Up: Visualizing Solutions

In the two previous lessons, visual representations such as number lines, bar models, hanger diagrams, balance scales, and drawings were used to represent equations.

1. Use the visual representation of your choice to represent the equation $37 = n + 12$.

   Answers will vary.

n	12
37	

2. Find the value of n.

   $n = 25$

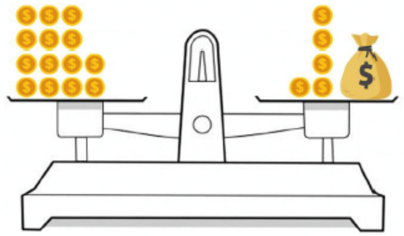
For students who have a good grasp of using inverse operations to solve one-step equations with addition or subtraction, challenge them to create a visual model for and solve a one-step equation using multiplication or division such as $6b = 42$ or $56 \div c = 7$.

14.4.2 Collaboration: Algebraic Equations From a Diagram

Component Overview: *Students collaborate to solve algebraic equations using the diagrams of a balance and a bar model. Students will draw on previous experiences with mathematical models and using inverse operations to solve one-step equations to complete this activity. Students may work with a partner to solve all the problems.*

14.4.2 Collaboration: Algebraic Equations from a Diagram

1. Ellen and Frey were given the balance scale shown below and told to write and solve an algebraic equation using inverse operations.



Ellen said, "There's no way to write an algebraic equation. If we can write an algebraic equation, we do not need inverse operations to solve it."

Frey said, "The only way to solve an algebraic equation is to use inverse operations to show the math is being reversed."

a. Whose statement do you agree with and why? Explain your answer to your partner.
Answers will vary. I agree with Frey's statement because inverse operations can be used to solve algebraic equations.

b. Write the algebraic equation represented by the balance scale.
$14 = 5 + x$

c. Solve the equation and show the steps you took.
$14 = 5 + x$
$14 - 5 = 5 + x - 5$
$9 = x$

d. How many gold coins are in the bag? Compare your answer with your partner.
9 *gold coins*

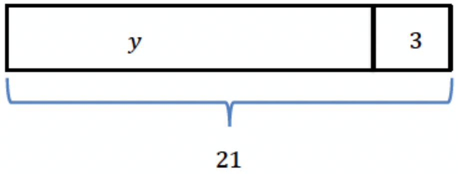
Use the vocabulary "algebraic equation" when discussing the activity, prompting students to give their ideas for what an algebraic equation can look like using references from what they have previously learned.

If you notice students are struggling to make the connection between the balance scale and algebraic equations, ask questions to help guide their thinking:

- *How do we know that the amounts on the scales are equal?*
- *How is a balance scale like an algebraic equation?*
- *Do you know how much money is in the bag?*
- *How could you represent the bag of money in an algebraic equation?*

14.4.2 Collaboration: Algebraic Equations From a Diagram (continued)

2. A bar model is shown.



a. Write an addition equation and a subtraction equation that can be represented by the bar model.

Addition Equation	Subtraction Equation
$y + 3 = 21$	$21 - 3 = y$

b. Describe the relationship between the two equations.
Answers will vary. The relationship between the two equations is that inverse operations can be used to rewrite one equation so that it is written like the other equation.

c. Find the value of y using either equation.

$y = 18$

d. Check your solution with your partner. Make any revisions if necessary.

$18 + 3 = 21$
$21 = 21$

Differentiation is built in throughout this activity. The balance scale and bar models provide visual entry points, allowing students to access the content through diagrams. Discussion is built in through partner work, providing additional auditory entry points. Consider having students use counters to model one or both problems. Students will physically move the counters to demonstrate performing the same action to both sides of an equation. This will provide an additional kinesthetic entry point.

Ensure students are making the connection between the visual models and the algebraic equations to build conceptual understanding. If you notice students are still struggling with the equations, ask questions like:

- *How does the addition equation relate to the bar model?*
- *How does the subtraction equation relate to the bar model?*
- *How are the equations related to each other?*
- *Why is the value of the variable the same in both equations?*
- *How can you check your solutions?*

14.4.3 Guided Practice: Solving One-Step Algebraic Equations Using Inverse Operations

Component Overview: *Students use inverse operations of addition and subtraction to solve for the missing value in a one-step equation. This activity builds on students' previous experiences with inverse operations.*

14.4.3 Guided Practice: Solving One-Step Algebraic Equations Using Inverse Operations

1. Use the equation $31 = m - 19$ to answer the following.

   a. What operation is being performed in the equation?
   Subtraction

   b. What is the inverse operation?
   Addition

   c. Solve the equation using the inverse operation to find the value of m. Show your work.
   $31 = m - 19$
   $31 + 19 = m - 19 + 19$
   $50 = m$

   d. Verify that the solution is correct by substituting the value found for m into the original equation.
   $31 = 50 - 19$

Review key vocabulary terms such as "equation, operation, inverse, value," and "substitute" with students. Have students point out examples of each in a mathematical problem. Encourage students to explain the terms in their own words and use precise mathematical vocabulary.

2. Use the equation $d + 64 = {}^{-}12$ to answer the following:

   a. What is the inverse operation that will be used to solve for d?
   Subtraction

   b. Write and solve the equation using the inverse operation.
   $d + 64 - 64 = -12 - 64$
   $d = -76$

   c. What is the value of d?
   $d = -76$

   d. Check your solution with your partner.
   $-76 + 64 = -12$
   $-12 = -12$

Consider asking the following questions to further scaffold instruction throughout this section:

- *What do you notice about the equation?*
- *What is the inverse operation? How does using the inverse operation balance the equation?*

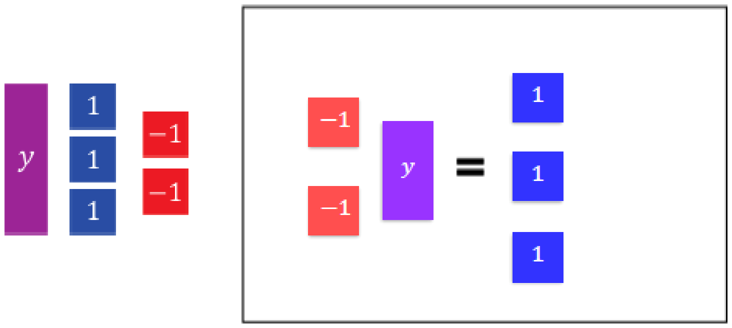
14.4.4 Exploration: Solving Equations Using Opposites

Component Overview: *Students use algebra tiles to model solving equations. Additionally, students explore solving an equation with a negative number. The understanding of opposites and inverses should develop to additive inverse. Encourage students to work with a partner and to discuss and explain each step of the process with each other.*

14.4.4 Exploration: Solving Equations Using Opposites

Work with your partner to complete the following.

1. Arrange the algebra tiles provided on the equation mat so that they represent the equation $-2 + y = 3$.



As students discuss zero pairs, they should be explaining the concept of additive inverse. Students do not need to name the property but by adding two positive tiles to solve the student is using the additive inverse. It may be helpful for some students to see this written $^{-}2 + 2 = 0$*. If students choose to use subtraction of negative two, they may struggle finding* $3 - {}^{-}2$ *because they struggle with operations with integers. This is why using the concept of zero pairs is used here. Continue to help students grasp the concept by using the terminology of zero pairs and additive inverse.*

2. Check your diagram with your partner.
   Answers will vary.

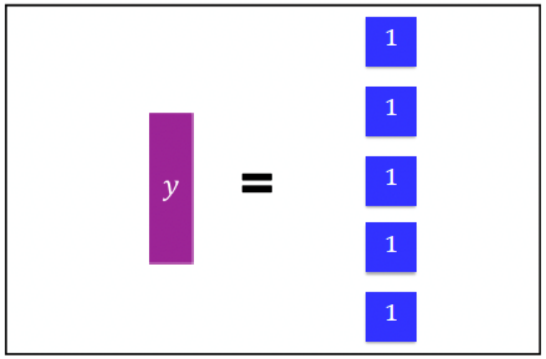
3. For this equation, there are no tiles that can be removed from both sides of the equation to find the value of y.
   Discuss with your partner how zero pairs could be used to find the value of y.
   Answers will vary. Zero pairs can be used by adding 2 "positive one" representations to both sides of the equation.

4. The goal when solving an equation is to isolate the variable. Which tiles can be added to both sides of the equation so that all tiles other than y on the left side of the equation can be eliminated?
   2 blue tiles

14.4.4 Exploration: Solving Equations Using Opposites (continued)

5. Explain why, when tiles are added to one side of the equation, they also must be added to the other side.
   Answers will vary. The same number of tiles must be added to both sides because the value on each side of the equation must remain equivalent and adding or subtracting the same amount on each side will maintain equivalency.

6. Draw tiles on the equation mat to represent the solution to the equation.



7. What is the value of y?
   $y = 5$

8. Discuss with your partner how solving for the variable in the equation $-2 + y = 3$, was different from solving for the variable in the equation $d + 64 = {}^{-}12$, even though both equations include addition. Summarize your discussion.
   Answers will vary. Solving for the variable for the equation $d + 64 = {}^{-}12$ involved using inverse operations with just numbers and solving for the variable for the equation $-2 + y = 3$ involved using inverse operations with modeling and zero pairs.

Consider having students use algebra tiles to model the problem. Students will physically move the tiles to demonstrate performing the same action to both sides of the equation. Have students verbally explain the process they are using to solve the equation.

This can lead to discussion about the fact that $-2 + y = 3$ and $y - 2 = 3$ are equivalent equations. When students are stuck on using inverse operations, they will often try to solve an equation like $-2 + y = 3$ by subtracting -2 from both sides of the equation because "it's the inverse operation." Students need to understand that inverse operations are helpful to isolate variables, but they can also consider the terms and use their knowledge of creating zero pairs to eliminate terms.

To provide additional scaffolding for this section, ask questions like:

- *What process do you use when solving an equation using zero pairs?*
- *What process do you use when solving an equation with inverse operations?*
- *How are these approaches similar? How are they different?*
- *Do you think one approach would be better to use in certain situations? When? Why?*

14.4.5 Collaboration: Solving Equations

Component Overview: *Students work with a partner to solve several one-step equations using addition or subtraction. Students can solve each equation with their partner, or each partner may solve one column of equations and then explain the process they used to their partner.*

14.4.5 Collaboration: Solving Equations

With your partner, determine who will be Partner A and who will be Partner B.

1. Solve the equations from your designated column.

Partner A	Partner B
$18 = 8 + x$ $18 - 8 = 8 - 8 + x$ $x = 10$	$-8 = -18 + x$ $-8 + 18 = -18 + 18 + x$ $x = 10$
$p - 9 = 16$ $p - 9 + 9 = 16 + 9$ $p = 25$	$9 = p - 16$ $9 + 16 = p - 16 + 16$ $p = 25$
$469 = 456 + w$ $469 - 456 = 456 - 456 + w$ $w = 13$	$-469 + w = -456$ $-469 + 469 + w = -456 + 469$ $w = 13$

2. The equations in each row have the same solutions. Check your work with your partner, making revisions to your work if necessary.

$18 = 8 + 10$ $\quad -8 = -18 + 10$

$25 - 9 = 16$ $\quad 9 = 25 - 16$

$469 = 456 + 13$ $\quad -469 + 13 = -456$

As you circulate listen for students using terminology such as zero pairs or additive inverse for the equations with negative numbers on the side with the variable.

For students who finish early ask them to write an explanation as to why each of the questions are the same answer. Look for students who understand that this distance between the values in the equations are the same.

Encourage students to explain the strategies they used to solve the equations. To support students, provide sentence frames such as, "I noticed that ______, so I ______." Or: "First, I ______ because ______."
When students share their answers with a partner, prompt them to rehearse what they will say when they share with the full group. Rehearsing provides opportunities for students to clarify their thinking.

14.4.6 Your Turn!

Component Overview: *Student's practice solving one-step equations using addition or subtraction. Students may use inverse operations or opposites to solve the equations. Allow students to use visual models as needed.*

14.4.6 Your Turn!

1. Solve each equation. Show your steps in finding the solution.

   a. $h + 37 = 62$

   $h + 37 - 37 = 62 - 37$

   $h = 25$

   b. $18 = y - 23$

   $18 + 23 = y - 23 + 23$

   $41 = y$

   $y = 41$

If students are struggling to solve the equations, ask questions like the following to provide further scaffolding:

- *What strategy can you use to solve the equation?*
- *How can you isolate the variable?*
- *What operation did you do to both sides of the equation?*
- *Why do you perform the same action on both sides of an equation?*

14.4.7 Wrap-Up: Cool Down

Component Overview: *Consider using the Wrap-Up as a formative assessment to gather data on student progress and vocabulary understanding from this lesson. This Wrap-Up assesses students' abilities to explain how to use inverse operations to solve a one-step equation using addition or subtraction.*

14.4.7 Wrap-Up: Cool Down

1. Your friend Giselle is absent today and will need the notes from class. Write a quick note to help her understand how to use inverse operations to solve the equation $25 = y + 7$.

   Answers will vary. To solve the equation using inverse operations, use the inverse of addition. Subtract 7 from both sides of the equation. Then check your work by substituting your solution into the original equation to check for equivalent values on both sides of the equal sign.

Students may also use terminology such as zero pairs or additive inverse.

For students who might struggle to represent their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.

14.5 Writing and Solving One-Step Addition and Subtraction Equations Within Real-World Contexts

Lesson Introduction

Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.	Learning Target	• I can write and solve one-step equations in one-variable within a real-world context using addition and subtraction.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• How do you write an algebraic equation to represent a real-world situation? • How do you know what the variable represents in a real-world situation?
Lesson Narrative	Students will work to create one-step equations using real-world contexts. The lesson starts with having students look at the variety of ways an equation can be written and progresses to having students write their own equations using real-world scenarios.

Component Summary	14.5.1 Warm-Up (~5 min.)	Create an equation based on a word problem *(SE p. 361)*	14.5.4 Collaboration (10-15 min.)	Represent word problems with equations *(SE p. 364)*
	14.5.2 Collaboration (5-10 min.)	Write and solve equations in real-world contexts *(SE p. 362)*	14.5.5 Your Turn! (5-10 min.)	Write and solve algebraic equations representing real-world scenarios *(SE p. 366)*
	14.5.3 Guided Practice (10-15 min.)	Create algebraic equations based on word problems *(SE p. 363)*	14.5.6 Wrap-Up (~5 min.)	Reflect on confidence in the lesson's learning target *(SE p. 366)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms:</u> • Variable • Equation	N/A	N/A

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may not realize that equations such as $x + 4 = 9$ are equivalent to $9 - x = 4$ and $9 - 4 = x$. • Students may believe words like "less than" can only be characterized with subtraction (for example, see 14.5.2 Collaboration, question 1).	• Determine if 14.5.2 Collaboration and/or 14.5.3 Guided Practice would be better as whole class, small group, or partner activities for your students. • If time is a problem, then consider skipping 14.5.4 Collaboration and/or assigning 14.5.5 Your Turn! for homework. • In the 14.5.4 Collaboration, students are creating their own word problem. This goes beyond the MA.6.AR.2.2 benchmark, but is doing so in an effort to help the student solidify their understanding of writing an equation from a context. Determine if this section will be helpful for your students or if this section needs be scaffolded further for your students. • Students who can do the mathematics in their head may struggle with the purpose of writing an equation. These students may also write the equations as $x = 24 - 8$ because that is how they see the problem. For these students it may be best to ask that they write two equivalent equations. Forcing the student to write $x + 8 = 24$ cannot be justified mathematically since their equation is equivalent. It may be helpful to have them state the expected form of the equations to be $x + p = q$ or $x - p = q$.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Collect and Display Prepare students for discussion by asking them to first share their responses to "Why are your solutions correct?" with a partner. Listen for, collect, and display vocabulary and include any diagrams that students use to represent each equation. Throughout the lesson, continue to update collected student language and remind students to borrow language from the display as needed. This will help students use mathematical language during their paired and whole-group discussions.	**MA.K12.MTR.7.1: Real-World Contexts** Students will be working to write and solve equations to represent real-world contexts. Clarifications within this standard emphasize connecting mathematical models to everyday experiences. Use this as an opportunity to provide students with practice determining if their equations are appropriate to solve the real-world problems described.

Differentiate Instruction	
	Consider placing students in groups and assigning groups to each number of the 14.5.4 Collaboration. By differentiating which groups engage with each portion of the activity, the collaboration can become a jigsaw in which students are responsible for sharing their learning with others. When students are positioned as experts in their respective field (or question), they internalize more agency and accountability in what they are presenting.

Lesson Notes:

14.5.1 Warm-Up: Bell Work

Component Overview: *Students write an equation to represent a real-world situation. Writing an equation draws on previous lessons in which students wrote expressions to represent situations.*

14.5.1 Warm-Up: Bell Work

1. Before last night, Faith had p Instagram posts on her account. She made 4 more posts last night. She now has 313 Instagram posts on her account. Write an equation that can be used to find p, the number of posts Faith had on her account before last night.

   $p + 4 = 313$

Students may write an expression such as $313 - 4$ or the equation $x = 313 - 4$. Students often struggle writing equations for contexts where they can easily determine the solution. Help students understand that they are actually using algebraic reasoning because they understand what the unknown is. Allow students to share whatever they wrote without considering what is right or wrong. Use this information to inform your teaching and support students as they learn to write equations.

14.5.2 Collaboration: Solving Equations in Real-World Contexts

Component Overview: *Students will be collaborating to select an equation to represent a situation, describe the unknown, and then solve the equation. All equations will have only one variable and use addition or subtraction. Students will draw on their previous experience using inverse operations to solve equations.*

14.5.2 Collaboration: Solving Equations in Real-World Contexts

Work with your partner to complete the following.

1. Kiran's cat weighs 5 pounds less than Clare's puppy. Clare's puppy weighs 14 pounds.
   a. Select all of the equations that could be used to find the weight of Kiran's cat.
      - ☐ $x + 5 = 14$
      - ☐ $x - 5 = 14$
      - ☐ $x = 14 + 5$
      - ☐ $x = 14 - 5$
      - ☐ $x - 14 = 5$
      - ☐ $x + 14 = 5$

   b. What does the variable, x, represent in this situation?
      The pounds that Kiran's cat weighs

   c. What is the weight of Kiran's cat?
      9 *pounds*

2. Kiegan collected 7 more items than his cousin Joel did for the annual toy drive. Joel collected 12 items.

   a. Select all of the equations that could be used to find the number of items Kiegan collected.
      - ☐ $t = 7 - 12$
      - ☐ $t - 7 = 12$
      - ☐ $t + 7 = 12$
      - ☐ $t = 12 + 7$
      - ☐ $t + 12 = 7$
      - ☐ $t - 12 = 7$

   b. What does the variable, t, represent in this situation?
      The number of items that Kiegan collected

Consider working with students to identify the key parts of the word problem. Have students highlight, circle, or underline the information that may be helpful when writing the equation.

"Kiran's cat weighs 5 pounds $\boxed{\text{less than}}$ Clare's puppy."

Students may incorrectly say that the variable t represents Kiegan instead of understanding the t represents the number of items Kiegan collects.

14.5.2 Collaboration: Solving Equations in Real-World Contexts (continued)

c. How many items did Kiegan collect for the toy drive?

19 items

3. Discuss with your partner what you notice about the equations that represent the two situations. Summarize your discussion.

Answers will vary. I notice that the equations can be written as addition or subtraction equations.

Encourage students to examine the process of writing an equation by asking the following questions:

- *Why could three equations represent the situation?*
- *How do each of the equations represent what is occurring in the problem?*

Specifically, have students make connections between the details of the real-world problem and the equation they select. Encourage students to manipulate each of the equations to see that they are equivalent.

14.5.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

Component Overview: *Students use real-world examples to explore writing an algebraic equation to represent a relationship, identifying the variable in the equation, and then solving the equation to answer a real-world question. This may be completed as a whole-group activity or with students working together in pairs or small groups.*

14.5.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

1. Jamal, a student at McLaughlin Middle School in Polk County, is preparing for a trip to Disney World with his mom and dad. An annual adult ticket is $\$139$. An annual child's ticket is $\$5$ less than an annual adult ticket. Jamal's parents are going to spend a total of $\$412$, before tax, on tickets.

   a. Write an algebraic equation to represent the relationship between the cost of a child's ticket and the cost of an adult ticket in this situation.
   Answers will vary. $139 - 5 = x$; $x + 5 = 139$

   b. What does the variable in your equation represent?
   The cost of a child's ticket

   c. Solve the equation.
   $x = 134$

   d. What was the cost of Jamal's ticket?
   $\$134.00$

2. After doing some additional research, Jamal's mom also decided to purchase the meal bundle for their trip. The meal plan price is different for adults than children. It costs $\$110$ for his mom and dad. The total for all three meal plans was $\$136$. An equation can be written to find the unknown cost of Jamal's meal plan.

   a. Write an algebraic equation that can be used to find the cost of Jamal's meal plan.
   Answers will vary. $110 + x = 136$; $136 - 110 = x$

   b. What does the variable in your equation represent?
   The cost of Jamal's meal plan

Consider allowing students to work on the 14.5.3 Guided Practice independently if they already have a firm grasp on writing and solving equations representing real-world situations.

Guide students through the connection of equivalent inverse equations that could represent either of the situations in questions 1 and 2. Remind students that they are solving to identify the unknown within each word problem and that variables represent the unknown numbers. Also work with students to identify keywords within the word problems.

14.5.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value (continued)

c. Solve the equation.
$x = 26$

d. How much was Jamal's meal plan?
$26.00

3. Use the equation $49 + t = 107$ to complete the following.

a. Write a word problem that can be represented by the equation $49 + t = 107$.
Answers will vary. Melinda sold 49 tickets so far and needs to sell t more to reach her goal of selling 107 tickets.

b. What does the variable represent based on the word problem you created?
Answers will vary. The number of tickets Melinda needs to sell to reach her goal.

c. What is the value of the variable?
$t = 58$

d. Share **only** your word problem with your partner. Use your partner's word problem to fill in the table.
Answers will vary.

Partner's Word Problem	What Does the Variable Represent?	Solve the Equation	Partner's Initials
If Cassey has read 49 pages of a book that has 107 pages, how many pages are left to read?	*The number of pages left to read*	$t = 58$	

Form a small group of students who need additional support. As you work with this group, consider reading the problem aloud to students and helping them identify and highlight key words and numbers in the problems. Assist students as they use the key words and numbers to construct and solve the equations. Encourage students to create and use visual aids, diagrams, and/or manipulatives as needed.

Scaffold the process of writing and solving a word problem for students as needed:

- *Select a real-life situation to write about.*
- *What does the 49 represent? What about 107?*
- *What does t represent?*
- *What inverse operation would you use to solve the equation?*
- *What does t equal?*
- *What does this represent in the word problem?*

14.5.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Component Overview: *Students collaborate to write equations to represent word problems and then use the equations to solve for an unknown in each problem. Students may work with a partner on each problem, or each can complete alternating sections of the problems as they progress.*

14.5.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Work with your partner to complete the following.

1. PJ and Jasmine were given the following word problem:

   Lexi's school is having a food drive. At the end of today's lunch period, Lexi collected a total of 34 *non-perishable items for the food drive. Lexi celebrated, as it was* 11 *more than yesterday's entire lunch collection! How many total non-perishable items were collected yesterday?*

Jasmine told PJ they needed to solve the problem using an addition equation. PJ disagreed and said they need to use a subtraction equation.

a. With your partner, determine who will follow Jasmine's suggestion and who will follow PJ's. Then, write the equation based on whose suggestion you are assigned and solve.
   Answers will vary.

Jasmine's Suggestion "Addition"	PJ's Suggestion "Subtraction"
$x + 11 = 34$ $x = 23$	$34 - 11 = x$ $x = 23$

b. Compare your equations and discuss your solutions with your partner.
   $23 + 11 = 34$
   $34 = 34$

c. Explain who was correct in their initial suggestion – Jasmine, PJ, both of them, or neither of them.
   Answers will vary. Both of them are correct because the equation can be written using addition or subtraction.

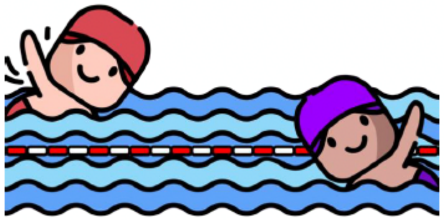
Consider assigning questions to groups of students based on readiness levels. Question 1 provides more scaffolding, as the word problem clearly lays out a scenario in which the equations will be exact. Question 2 may be more difficult for some students to access, as it requires creation of a scenario from which they must then also create an equation. Once the problems have been completed, engage the class in a jigsaw routine so that students can present their work to the class.

Throughout the 14.5.4 Collaboration, use the Collect and Display routine. Have students explain their reasoning for each question to their partner. Listen for, collect, and display vocabulary and include any diagrams that students use to represent each equation. Throughout the lesson, continue to update collected student language and remind students to borrow language from the display as needed.

14.5.4 Collaboration: Writing and Solving Equations for Real-World Contexts (continued)

d. How many non-perishable food items were collected during Lexi's lunch period yesterday?
23 non-perishable food items

2. An image of two swimmers in a pool is shown.



a. Work with your partner to create an addition or subtraction word problem based on the image.
Answers will vary. Swimmer 1 swam 3 more laps than Swimmer 2. If Swimmer 2 swam 9 laps, how many laps did Swimmer 1 swim?

b. Write an equation to represent your scenario.
Answers will vary. $3 + 9 = x$

c. What does the variable in your equation represent?
Answers will vary. The number of laps Swimmer 1 swam.

d. What is the value of the variable?
Answers will vary. $x = 12$

Encourage students to explain their reasoning by asking the following questions:

- *Why would you use an addition equation to represent the problem?*
- *Why would you use a subtraction equation to represent the problem?*
- *What does the variable represent in the equations?*

Challenge students to create a question in which a negative number must be used in the equation.

Encourage students to examine the equation by asking the following questions:

- *How does the equation relate to the word problem you wrote?*
- *How do the numbers in your equation represent what is described in your word problem?*
- *How can you use the equation to figure out the unknown?*

14.5.5 Your Turn!

Component Overview: *Students practice using addition and subtraction inversely in real-world scenarios. Students write an equation to represent each word problem and then use the equation to solve for the unknown.*

14.5.5 Your Turn!

Write and solve an algebraic equation for each of the following.

1. Priya and Gerri are working on an extra-credit poster for math class. Priya has 12 double-tip markers, which is 3 fewer than Gerri has. How many markers does Gerri have?

   $x - 3 = 12$
   $x = 15$

2. The most popular elective at Arts Academy is the improv class. Out of the 97 students that have applied, only 45 can be accepted. How many students are not accepted?

   $97 - 45 = x$
   $x = 52$

If students need support, consider allowing them to work with a partner. For students who need language support, read the word problems aloud to them. Encourage students to create diagrams or use manipulatives as needed.

14.5.6 Wrap-Up: Reflection

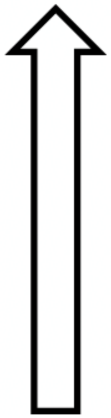
Component Overview: *Use this exercise as an opportunity to gather feedback on student progress from this lesson. This Wrap-Up assesses students' feelings about writing and solving one-step equations using addition or subtraction to solve real-world problems.*

14.5.6 Wrap-Up: Reflection

1. Shade in the arrow up to the statement that best describes your current understanding of the benchmark below:

   Answers will vary.

   Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



I'm so confident, I could explain how to write and solve one-step equations in one variable using addition or subtraction to someone else!

I can get to the right answer, but I don't understand well enough to explain how to write and solve one-step equations in one variable yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.

Consider other resources for additional enrichment opportunities for students to practice or test their abilities.

14.6 Solving One-Step Multiplication and Division Equations in One Variable Using Models

Lesson Introduction

Benchmark	MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.	Learning Targets	• I can solve one-step multiplication and division equations using manipulatives. • I can solve one-step multiplication and division equations using diagrams.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• How would you use algebra tiles to model and solve the multiplication and division equations? • How do you use tape diagrams to model and solve one-step multiplication and division equations? • How do you use hanger diagrams to model and solve one-step multiplication and division equations? • How do you solve multiplication and division one-step equations?
Lesson Narrative	In this lesson, students use manipulatives, tape diagrams, and algebra tiles to solve multiplication and division one-step equations using inverse operations. Students use inverse operations to isolate the variable while making sense of the meaning of the variable and the operation using models.

Component Summary	14.6.1 Warm-Up (~5 min.)	Error analysis of image *(SE p. 367)*	14.6.3 Guided Instruction (10-15 min.)	Learn how to model and solve division one-step equations *(SE p. 370)*
	14.6.2 Exploration (10-15 min.)	Explore how to model and solve multiplication one-step equations *(SE p. 368)*	14.6.4 Collaboration (10-15 min.)	Match one-step multiplication and division equations and solutions *(SE p. 372)*
	14.6.6 Wrap-Up (~5 min.)	Write a sales pitch for inverse operations *(SE p. 373)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Inverse operations	• Algebra tiles • Colored pencils • (Optional) Chart paper • (Optional) Markers	• Consider using virtual manipulatives. • Consider putting algebra tiles in zip top bags with the contents labeled for students to use or share. • If you do not have algebra tiles, you can create them using a template or use virtual algebra tiles.

Teacher Tips

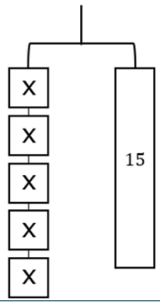
Common Misconceptions	Things to Consider
• Students may think that the coefficient represents the tens place and the variable represents the ones place (of a two-digit number). • Students may think the inverse operation of a negative coefficient is to add the coefficient to both sides (instead of dividing by it). • Students may not perform the inverse operation to both sides of the equation. • Students may struggle with the models for division equations.	• Saying "2 multiplied by x" instead of "$2x$" will help students remember that the coefficient is multiplied by the variable. • If time is an issue consider assigning Your Turn! as homework. • The models for division equations may be difficult for students. Consider have strips of paper that students can divide up into parts. For example, for 14.6.3 Guided Instruction question 1b provide the students with a strip of paper that is partitioned into 3 parts. Students can write $\frac{x}{3}$ on each part and cut the strip. Then the student can use the paper with $\frac{x}{3}$ to set equal to 4 and repeat that to demonstrate the model in question 1b.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Take Turns In 14.6.4 Collaboration, students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.	**MA.K12.MTR.2.1: Multiple Representations** In 14.6.2 Exploration, students learn how to model and solve multiplication one-step equations using algebra tiles. In 14.6.3 Guided Instruction, students learn how to model and solve division one-step equations using algebra tiles. After learning how to represent and solve one-step equations using algebra tiles, students practice in the subsequent components.

Differentiate Instruction

Visual, auditory, and kinesthetic entry points are built in throughout this lesson. Hanger diagrams can be physical or digital manipulatives that benefit students who learn best with hands-on resources.

Provide English Language Learners with anchors of support, such as vocabulary cards of the words used in this lesson: equation, positive, negative, inverse operations, integers, solutions.



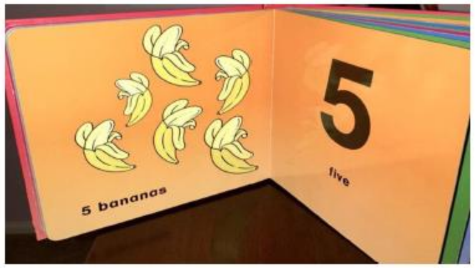
Lesson Notes:

14.6.1 Warm-Up: Math Fail

Component Overview: *Students analyze an image to determine the "math fail."*

14.6.1 Warm-Up: Math Fail

This children's book has published a "math fail."



1. Explain the error shown in this image.

   There are 6 bananas in the image, but the writing shows there should be five.

Consider an extension for homework in which students have to find "math fails" in their world—at the grocery store, on a sale sign, or in a book.

14.6.2 Exploration: Exploring Relationships

Component Overview: *Students explore how to model and solve multiplication one-step equations using algebra tiles. Students develop a conceptual understanding of equations as balanced scales. This helps students understand why they need to perform inverse operations to both sides of the equation. Finally, students review the inverse operations for addition, subtraction, multiplication, and division.*

14.6.2 Exploration: Exploring Relationships

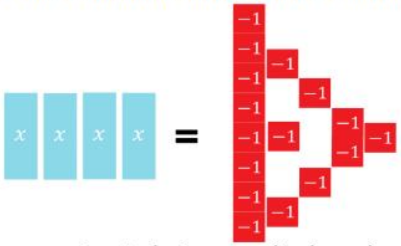
Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. A model of the equation $4x = -16$ is shown using algebra tiles.

   The 

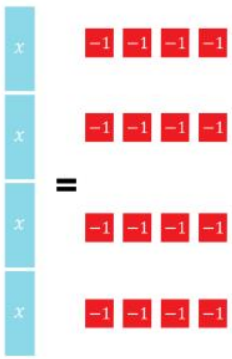
tile represents the variable, x, and the 

represents the constant term where each tile is 1 unit.



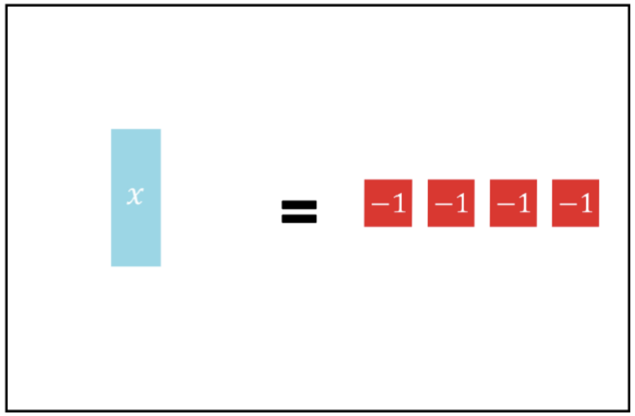
   a. What operation is being applied to the 4 and the x in the equation $4x = -16$?
   Multiplication

   b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?
   4 groups



Ask students how they think the variable can be isolated. This could be rephrased as grouping each side to show how many each x represents. Have students share their ideas and then test them using the algebra tiles. Students may be able to figure out how to solve the equation independently if given time to experiment with the algebra tiles.

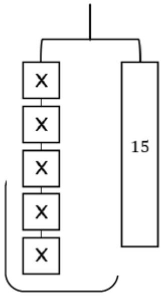
c. Use algebra tiles to sketch the solution of the equation.



d. What is the value of x?
$x = -4$

This section provides a variety of entry points through the visual representations of the hanger diagrams and partner discussions. Consider providing algebra tiles for a kinesthetic entry point and encouraging students to model the equation first using tiles if they are struggling with the hanger diagram representation.

2. The hanger diagram represents the equation $5x = 15$.



a. Complete the statements.

The left side of the hanger is broken into equal parts. If 15 is broken into the same number of equal parts, each part will have the same weight as one x block. Splitting each side of the hanger into

- ○ 3
- ○ 5
- ○ 15

the same as

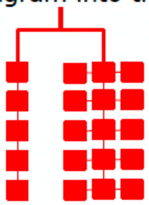
- ○ dividing
- ○ multiplying

each side of the equation by

- ○ 3.
- ○ 5.
- ○ 15.

14.6.2 Exploration: Exploring Relationships (continued)

b. Sketch splitting the hanger diagram into the equal parts as described in Part A.
Answers will vary.



c. What is the value of x?
$x = 3$

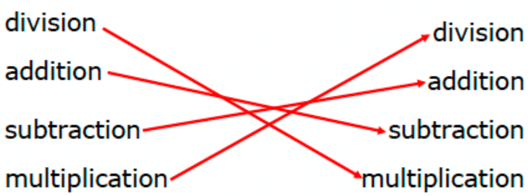
d. Compare your diagram with your partner. Discuss the similarities and differences.
Answers will vary.

e. Provide positive feedback to your partner in their book on their sketch and solution.
Answers will vary.

Partner Feedback	

3. Inverse operations can be helpful in solving equations.

a. Draw an arrow to match each operation on the left with its inverse operation on the right.



b. Explain what it means for operations to be inverses.
Answers will vary. Inverse operations "undo" each other.

c. Which inverse operations were used to solve the equations in this exploration?
Multiplication and division

Model what positive feedback looks and sounds like prior to this activity with examples. Consider providing a posted anchor chart that students could reference.

If students are struggling to match, explain inverse operations by further scaffolding question 3 through questioning:

- *What does "inverse" mean?*
- *How can you determine the inverse of an operation?*

14.6.3 Guided Instruction: Using Models to Solve Equations

Component Overview*: Students model and solve division one-step equations using algebra tiles. Students solve the equation by multiplying both sides of the equation by the number the variable was divided by. Remind students that they can check their solution by substituting it in for the variable and seeing if the left side equals the right side.*

14.6.3 Guided Instruction: Using Models to Solve Equations

Algebra tiles can also be used to model division equations.

1. The equation $\frac{x}{3} = 4$ can be rewritten as $x \div 3 = 4$.

   To represent division, show that x tile is being divided into equal parts. Sketch lines on the x tile to represent the fraction $x \div 3$.

   This means when x is divided into 3 parts, each part is equal to 4. The model shown illustrates that $\frac{x}{3} = 4$.

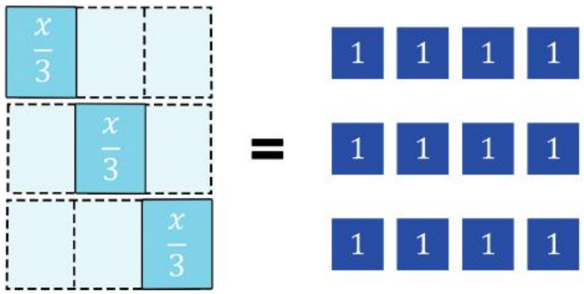
   a. If each $\frac{1}{3}$ of x is equal to 4, discuss with your partner what the value of x is. As part of your discussion, include which operation can be used to determine the value of x. Summarize your discussion.
   Answers will vary. x is equal to 12. The problem is using division so the inverse that would need to be used is multiplication.

Consider creating an anchor chart with instructions, examples, and visual models for students to reference in subsequent components.

Students could say addition or multiplication. If students say addition, it would be repeated addition, which can be related to multiplication.

14.6.3 Guided Instruction: Using Models to Solve Equations (continued)

b. The image models the relationship discussed that can be used to determine the value of x.



Explain how this representation can be used to determine the value of x .

Answers will vary.
One-third of the boxes is four.

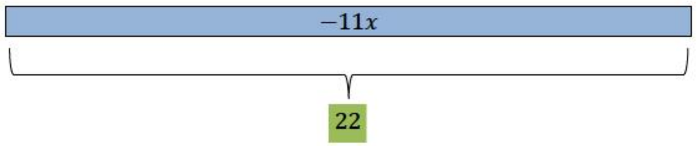
c. What is the value of x?
$x = 12$

d. Complete the statements by writing operations on each blank.

The original equation involved the operation *Division*. Its inverse operation, *Multiplication*, was used to solve the equation.

Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

2. The tape diagram models the equation $22 = -11x$.



Take the time to explain the model in question 1b. Point out how each $\frac{x}{3} = 4$. Then, help students see that the left side is three $\frac{x}{3}$'s which is equivalent to x, and since the left side has three $\frac{x}{3}$'s, the right side needs three sets of the four one-unit tiles. Lead students so that they relate the model to multiplication.

To provide additional scaffolding, ask guiding questions like:

- *What is the inverse operation of division?*
- *What is the inverse operation of multiplication?*
- *Why do you perform the inverse operations on both sides of the equation?*
- *How can you check your equation to make sure the solution is correct?*

14.6.3 Guided Instruction: Using Models to Solve Equations (continued)

The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1

$$\underbrace{\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}\boxed{-x}}_{22}$$

a. How were the integers in the equation used to determine the negative value of x?

Explanation of Step 1
Answers will vary. $-11x$ has been broken up into 11 individual pieces with each piece equivalent to $-x$.

Step 2

$$\underbrace{\boxed{-x}}_{2}$$

Explanation of Step 2
Answers will vary. The value of $-x$ is 2.

Step 3

$$\underbrace{\boxed{x}}_{-2}$$

Explanation of Step 3
Answers will vary. We want to find the value of x, so we take the opposite $-x$ and the opposite of 2, which gives us -2.

b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.
Answers will vary.

Consider providing sentence stems for students to discuss their responses and explanations to provide additional entry points for learners.
“The operation used in the equation was ___________. I solved the equation by using ___________. This worked because…”

If students are able to solve the equations without visual models, allow them to do so. Ask them to check their solutions. If they finish early, consider having them write real-world word problems that could be modeled by the equations in the lesson.

To encourage mathematical reasoning, ask questions like:

- *Does this answer make sense?*
- *Did you check your solution by substituting it back into the original equation?*
- *How can you use mental math to help you find the correct diagram that models the equation?*

14.6.4 Collaboration: Matching Equations and Solutions

Component Overview: *Students match one-step multiplication and division equations and solutions. Then students explain how they determined which equations to match with which solutions to their partners.*

14.6.4 Collaboration: Matching Equations and Solutions

1. Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

1. $-3x = -6$
2. $-\frac{x}{2} = 4$
3. $\frac{x}{3} = 3$
4. $-\frac{x}{2} = 5$
5. $3x = 9$
6. $\frac{x}{10} = -2$
7. $4x = 8$

Equation	Solution	Explanation for Match
1	*A*	*Answers will vary. When substituting 2 in for x in $-3x = -6$ the result is -6.*
2	*G*	*Answers will vary. When substituting -8 in for x in $-\frac{x}{2} = 4$ the result is 4.*
3	*D*	*Answers will vary. When substituting 9 in for x in $\frac{x}{3} = 3$ the result is 3.*
4	*C*	*Answers will vary. When substituting -10 in for x in $-\frac{x}{2} = 5$ the result is 5.*
5	*E*	*Answers will vary. When substituting 3 in for x in $3x = 9$ the result is 9.*
6	*B*	*Answers will vary. When substituting -20 in for x in $\frac{x}{10} = -2$ the result is -2.*
7	*A*	*Answers will vary. When substituting 2 in for x in $4x = 8$ the result is 8.*

Take Turns
Students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.

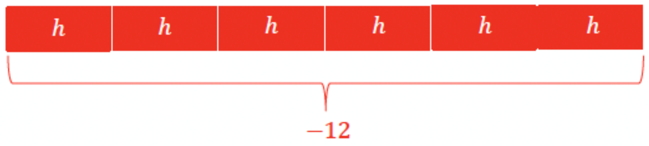
14.6.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students use a visual model to solve a one-step multiplication equation. Use data from this formative assessment to determine which students need remediation before the next lesson.*

14.6.5 Wrap-Up: Ticket Out the Door

1. Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.

   Diagram answers will vary. $h = -2$



This section allows for multiple entry points, as students have a choice for how they want to represent and solve the equation. Encourage students to choose the method that makes the most sense to them.

14.7 Solving One-Step Multiplication and Division Equations Algebraically

Lesson Introduction

Benchmark	MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.		Learning Target	• I can solve one-step multiplication and division equations in one-variable algebraically.
Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
Essential Question	• How can you use inverse operations to isolate the variable to solve one-step equations algebraically?			
Lesson Narrative	In this lesson, students will solve multiplication and division one-step equations algebraically using inverse operations. Students begin this lesson with an activity that allows them to explore how performing inverse operations when trying to find the solution of a variable in an equation helps to isolate the variable and keep the equation balanced. Students then use inverse operations algebraically to isolate the variable and find a solution.			
Component Summary	14.7.1 Warm-Up (~5 min.)	Write equations with a given solution *(SE p. 374)*	14.7.3 Guided Practice (10-15 min.)	Solve one-step multiplication and division equations algebraically *(SE p. 376)*
	14.7.2 Guided Instruction (10-15 min.)	Solve equations using inverse operations *(SE p. 375)*	14.7.4 Your Turn! (10-15 min.)	Practice solving one-step multiplication and division equations algebraically *(SE p. 377)*
	14.7.5 Wrap-Up (~5 min.)	Explain and correct an error in a sample of student work *(SE p. 378)*		
Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	N/A	• Consider creating an anchor chart with instructions, examples, and visual models.	

Teacher Tips

Common Misconceptions	Things to Consider
• Students may believe that since $7x$ divided by 7 results in x that the coefficient of x is zero instead of one. • Students may not perform the inverse operation to both sides of the equation. • Students may not know what to do when the variable is on the right side of the equal sign.	• During 14.7.2 Guided Instruction, it is important to link the algebraic methods to the visual and concrete methods previously used. • During 14.7.2 Guided Instruction, explain what the equal sign means to the equation and how it keeps both sides of the equation balanced. • For question 4 in 14.7.2 Guided Instruction, tell students that the variable can be on either side of the equation. If they prefer, demonstrate how to rewrite the equation with the variable on the left side. Explain that this is the symmetric property of equality, if $a = b$ then $b = a$.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Take Turns In 14.7.3 Guided Practice, students explain the rationale for their solutions and listen to their partner's rationales. Taking turns deciding, explaining, and listening limits one student taking over and the other not participating. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.	**MA.K12.MTR.4.1: Discussion and Reflection** In the 14.7.5 Wrap-Up, students identify and correct errors in a sample of student work. Students write an explanation of the error and describe how they would correct it. This activity will help students avoid making similar errors themselves in the future.

Differentiate Instruction

14.7.4 Your Turn! involves students working independently. If students struggle solving equations algebraically, provide them with concrete manipulatives or tell them to draw visual models. Also, consider creating and displaying an anchor chart with instructions, examples, and visual models to reference. Finally, if students struggle with their multiplication and division facts, allow them to use a multiplication chart until they become fluent in these facts.

Lesson Notes:

14.7.1 Warm-Up: Different Problems, Same Solution

Component Overview: *Students write four equations that all have a solution of $x = 3$. The directions do not specify what type of equations students should write. Clarify that students should write one-step multiplication and division equations. Consider adding additional requirements, such as two multiplication and two division equations or requiring each equation to include a negative integer.*

14.7.1 Warm-Up: Different Problems, Same Solution

1. Write four equations whose solutions are all $x = 3$.

   Answers will vary. $6x = 18$; $4x = 12$; $12x = 36$; $\frac{x}{3} = 1$

If students finish early, challenge them to write a real-world problem that can be modeled by each of their equations.

14.7.2 Guided Instruction: Using Inverses to Solve Equations

Component Overview: *Students work with a partner to solve equations using inverse operations. Consider creating an anchor chart and displaying it for students to reference in subsequent components. Explain the connections between using inverses and using visual models.*

14.7.2 Guided Instruction: Using Inverses to Solve Equations

Inverse operations are used to solve equations by isolating a variable of interest.

When solving one-step equations using visual models, the inverse operation performed on one side of the model is also performed on the other side to keep the equation balanced.

1. With your partner, re-visit the meaning of balanced equations. Write a brief summary explaining what it means for equations to be balanced and how balance is maintained when solving. Include examples as part of your summary.
   Answers will vary. For an equation to remain balanced, inverse operations will need to be performed on both sides of the equal sign.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Multiplication (×)	*Division* (÷)
Division (÷)	*Multiplication* (×)

Explain the meaning of the equal sign and how it keeps both sides of the equation balanced. Remind students of the balance scale they saw in previous lessons. Explicitly link the algebraic methods to the visual and concrete methods previously used.

14.7.2 Guided Instruction: Using Inverses to Solve Equations (continued)

3. Consider the equation $7m = 21$ to complete the following.

   a. Discuss with your partner how to solve $7m = 21$.
   Answers will vary. In the equation, $7m$ means 7 times m. The inverse of multiplication is division, so you should divide both sides of the equation by 7 to solve for m.

   b. What operation is occurring between the variable m and its coefficient of 7?
   Multiplication

   c. To isolate the variable m from its coefficient of 7, what inverse operation should be performed?
   Division

   d. Work with your partner to determine the value of m.
   $m = 3$

4. Consider the equation $13 = \frac{h}{-5}$ to complete the following:

   a. What mathematical operation is represented by $\frac{h}{-5}$?
   Division

   b. What is the inverse operation?
   Multiplication

   c. Perform the inverse operation in the boxes below.

   $$\boxed{-5}\,(13) = \left(\frac{h}{-5}\right)\,\boxed{(-5)}$$

   d. What is the value of h? $h = -65$

   e. Discuss with your partner how to check whether or not your solution is correct. Then, check your solution. *Answers will vary. Substitute -65 in for h. Divide -65 by -5 and the result will equal 13.*

After solving for m, facilitate a discussion regarding what happened when the coefficient 7 was divided by 7.

It's important to distinguish that the coefficient of m becomes 1 in this case, not 0. After solving addition and subtraction equations, students come to understand that the constant term is eliminated when zero pairs are made. The term being added or subtracted becomes 0 when solving.

For multiplication and division, this is not the case. It's important to make this distinction and develop this understanding in 6th grade when students are working with one-step equations to develop strong foundations as students become more flexible in future courses where they'll learn to solve two-step, multi-step, and literal equations.

To encourage mathematical reasoning, ask questions like:

- *Does this answer make sense?*
- *How did you decide the answer made sense?*
- *How can you determine if the answer will be positive or negative?*

14.7.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

Component Overview: *Students collaboratively practice solving one-step multiplication and division equations algebraically. Then students use substitution to check their solutions. Circulate while students work to monitor progress and decide which students you will call on to share their solutions during the whole-class debrief.*

14.7.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

1. Complete the table.

Equation	Description of How to Solve	Work	Solution
$6x = 18$	Divide both sides of the equation by 6.	$\frac{6x}{6} = \frac{18}{6}$	$x = 3$
$\frac{h}{5} = 7$	*Multiply both sides of the equation by 5.*	$(5)\frac{h}{5} = 7(5)$	$h = 35$
$-36 = \frac{m}{-6}$	*Multiply both sides of the equation by -6.*	$(-6)(-36) = \left(\frac{m}{-6}\right)(-6)$	$m = 216$
$-32 = 8y$	*Divide both sides of the equation by 8.*	$\frac{-32}{8} = \frac{8y}{8}$	$y = -4$
$4 = -\frac{b}{12}$	*Multiply both sides of the equation by -12.*	$(-12)(4) = \left(-\frac{b}{12}\right)(-12)$	$b = -48$

Take Turns
Students explain the rationale for their solutions and listen to their partner's rationale. Taking turns deciding, explaining, and listening limits one student taking over and the other not participating. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.

14.7.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically (continued)

2. With your partner, verify that each solution is correct using substitution.

$(6)(3) = 18$
$18 = 18$

$\frac{35}{5} = 7$
$7 = 7$

$-36 = \frac{216}{-6}$
$-36 = -36$

$-32 = 8(-4)$
$-32 = -32$

$4 = -\frac{-48}{12}$
$4 = 4$

The partner discussions about the correctness of their solutions provide an auditory entry point for students to discuss their methods and solutions.

14.7.4 Your Turn!

Component Overview: *Students independently practice solving one-step multiplication and division equations algebraically. Students may find it helpful to reference an anchor chart with instructions, examples, and visual models.*

14.7.4 Your Turn!

Solve each equation algebraically. Show your work.

1. $\frac{a}{4} = 10$ $\frac{a}{4}(4) = 10(4)$ $a = 40$	2. $-11 = \frac{x}{8}$ $(8)(-11) = \frac{x}{8}(8)$ $x = -88$
3. $-12p = -84$ $\frac{-12p}{-12} = \frac{-84}{-12}$ $p = 7$	4. $-42 = 6x$ $\frac{-42}{6} = \frac{6x}{6}$ $x = -7$
5. $18c = 72$ $\frac{18c}{18} = \frac{72}{18}$ $c = 4$	6. $\frac{r}{6} = 36$ $\frac{r}{6}(6) = 36(6)$ $r = 216$

If students struggle to solve equations algebraically, provide them with concrete manipulatives or encourage them to draw visual models. This will provide additional kinesthetic and visual entry points as students work to solve independently.

14.7.5: Wrap-Up: Error Analysis

Component Overview: *Students explain and correct an error in a sample of student work. Use data from this formative assessment to determine which students need remediation before the next lesson.*

14.7.5 Wrap-Up: Error Analysis

1. While checking her homework from last night, Lola noticed she got the following problem incorrect. Examine Lola's work and write a note to her detailing how she can correct her mistake.

$$\begin{array}{r} 42 = h \div 7 \\ +7 \qquad +7 \\ \hline 49 = h \end{array}$$

Answers will vary. Lola added 7 to both sides of the equation instead of multiplying. The inverse of division is multiplication. If Lola would have multiplied both sides by 7 she would have gotten the problem correct. $h = 294$

Consider providing students with sentence stems or answer choices if they struggle to begin writing their responses. This activity will help students avoid making similar errors themselves in the future.

14.8 Writing and Solving One-Step Equations in One Variable

Lesson Introduction

Benchmarks	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.	Learning Targets	• I can write and solve one-step equations in one variable within a real-world context using multiplication and division. • I can solve one-step equations in one variable using addition and subtraction.	
Benchmark Coherence	Building On	Working Toward		
	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.		
Essential Questions	• How do you write and solve multiplication and division one-step equations from a real-world context? • How can you check your solution to an equation is reasonable?			
Lesson Narrative	In this lesson, students connect one-step multiplication and division equations to real-world situations. Students are first presented with real-world situations and find the equation that models each situation. Students are given real-world situations where they have to write an equation to represent the situation and solve the equation. The lesson ends with a mix review of solving one-step equations.			
Component Summary	14.8.1 Warm-Up (~5 min.)	Solve a puzzle of one-step equations *(SE p. 379)*	14.8.4 Guided Practice (10 min.)	Write and solve one-step real-world multiplication and division equations *(SE p. 381)*
	14.8.2 Collaboration (5 min.)	Collaboratively match real-world situations to one-step equations *(SE p. 379)*	14.8.5 Try It (10 min.)	Mixed review writing and solving one-step equations *(SE p. 382)*
	14.8.3 Guided Instruction (15 min.)	Write and solve one-step real-world multiplication and division equations *(SE p. 380)*	14.8.6 Wrap-Up (~5 min.)	Self-assessment of mastery of writing and solving one-step equations *(SE p. 383)*
Resources	Glossary	Materials	Additional Preparation	
	Key Terms: N/A Additional Terms: N/A	• (Optional) Chart paper • (Optional) Markers	• Consider modifying the context of some questions to connect them to contexts students in your class are interested in. • Consider including an example problem for each learning target in the 14.8.6 Wrap-Up to help students accurately self-assess their mastery of the target.	

Teacher Tips

Common Misconceptions	Things to Consider
• Students may not perform the inverse operation on both sides of the equation. • Students may believe each situation can only be represented by one equation, therefore struggling to identify equivalent equations.	• Have students continue to create their own word problems using different equations and verbalize what the variable is in their problem. • If students are having problems with creating their own equations for the questions, have them draw a visual model that represents the problem and then translate the visual model into an equation. • If time is an issue, then consider assigning 14.8.5 Try It for homework. • Students who can do the mathematics in their head may struggle with the purpose of writing an equation. These students may also write the equations as $x = 24 \div 8$ because that is how they see the problem. Their equation could be mathematically accurate and could not be considered wrong. For these students it may be best to ask that they write two equivalent equations. Forcing the student to write $8x = 24$ cannot be justified mathematically since their equation is equivalent. It may be helpful to have state the expected form of the equations to be $px = q$ or $\frac{x}{p} = q$.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share In 14.8.2 Collaboration, ask students to initially determine the matches that they believe are correct independently. Then have students discuss their answers and justifications with a partner. These discussions will build student confidence and understanding.	**MA.K12.MTR.7.1: Real-World Contexts** In 14.8.3 Guided Instruction, students write and solve one-step multiplication and division equations based on real-world situations. These questions will help students develop an appreciation of the utility of math. Students will be increasingly more engaged if they are connected or interested in the context.

Differentiate Instruction

The scaffolding support of visual models have been used throughout this unit but are not prominent in this lesson. Some students may still require the support of the visual models as they work to solve one-step multiplication and division equations. Encourage students who may benefit from additional visual entry points to draw bar models before solving. Students may benefit from the additional kinesthetic entry point of algebra tiles as well. Consider providing these scaffolds during the 14.8.3 Guided Instruction and 14.8.4 Guided Practice portions of this lesson and releasing students to solve the rest of the problems algebraically if they can.

Lesson Notes:

14.8.1 Warm-Up: One-Step Equations

Component Overview: _Students create true equations by filling in the empty boxes using the digits one through nine._

14.8.1 Warm-Up: One-Step Equations

1. Fill in each blue box to make the equations true. Use only digits 1 to 9, at most one time each.

$$\boxed{4} + a = \boxed{9}$$

$$\boxed{3}\, b = \boxed{6}$$

$$c - \boxed{7} = \boxed{1}$$

$$a = \boxed{5},\ b = \boxed{2},\ c = \boxed{8}$$

Consider having students share their strategies for filling in the boxes so that each equation is true.

14.8.2 Collaboration: Relating Real-World Contexts to Equations

Component Overview: *Students collaboratively match real-world situations to one-step equations.*

14.8.2 Collaboration: Relating Real-World Contexts to Equations

1. Four equations are shown.

$$4x = 32 \qquad 32x = 4$$
$$\frac{x}{32} = 4 \qquad \frac{x}{4} = 32$$

Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity. Then, write the matching equation in the space provided.
Answers will vary.

Situation 1	Situation 2
At a restaurant, 4 friends decided to split the bill evenly. If each person paid \$32, what was the total bill?	A manatee has a total of 32 teeth. If a manatee has 4 rows of teeth with the same number of teeth in each row, how many teeth are in each row?
Equation: $\frac{x}{4} = 32$	**Equation:** $4x = 32$

Think-Pair-Share
Consider structuring this activity with a Think-Pair-Share routine.

- *Think: Students independently read each situation and determine the matching equation.*
- *Pair: Students turn to a partner to justify answers and discuss any differences.*
- *Share: Prompt students to share whole group any reasoning from their partner that helped to change their approach or answer.*

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

Component Overview: *In 14.8.3 Guided Instruction, students begin by creating a bar model to represent a real-world scenario. Students then create an equation to represent the model and use it to solve for the unknown variable, explaining what the solution means in the context of the original problem. Students then work with a partner to determine which equations accurately represent another scenario and choose an equation to solve for the unknown. Students then check their work using a different equation and explain their solution in the context of the original problem.*

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

1. A town's total budget for firefighters' wages is $320,000. Wages are calculated at $40,000 per firefighter.

   a. Draw a bar model that can be used to determine how many firefighters the town can afford to employ.
   Answers will vary.

   b. Write an equation that can be used to find w, the number of firefighters the town can employ.
   $40{,}000w = 320{,}000$

   c. What is the value of w?
   $w = 8$

   d. Explain what the solution means in the context of the problem.
   Answers will vary. The town can hire 8 firefighters.

Remind students that they can create visual models for all questions (even if the directions do not require them to). Also, model how to check the solution and remind students to check their solutions.

MA.K12.MTR.7.1: Real-World Contexts
Students write and solve one-step multiplication and division equations based on real-world situations. These problems will help students develop an appreciation of the utility of math while providing meaningful engagement for making sense in familiar contexts.

Consider allowing students to adjust the context based on their own lived experiences. For example, question 1 could be adjusted to school budgets and spending per student for meaningful conversations on equitable school funding, or question 2 could be adjusted to be music samples and recording studios in town.

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division (continued)

2. There are 4 flooring companies submitting bids on a new housing development. Each company has 11 different flooring samples.

   Dino said, "To find the total number of samples, s, being submitted, set up the equation $s = 4 \cdot 11$." Dino's partner Callahan shared that he used the equation $\frac{s}{4} = 11$ to find the total number of samples submitted.

   Bristol shouted across the classroom that she wrote the equation, $\frac{s}{11} = 4$.

   a. Discuss with your partner which student — Dino, Callahan, or Bristol — wrote the correct equation to solve for the total number of samples submitted.
   Answers will vary. All three wrote an equation that can be solved to find the number of samples.

   b. Solve one of the equations and then use a different equation to check your work.
   Answers will vary.
   Dino: $s = 4 \cdot 11$; $s = 44$
   Callahan: $\frac{s}{4} = 11$; $\frac{44}{4} = 11$
   Bristol: $\frac{s}{11} = 4$; $\frac{44}{11} = 4$

   c. Explain how the answer makes sense in terms of the context.
   Answers will vary. There are 44 samples. Multiplication and division are inverse operations therefore all three students are correct. Four stores times 11 samples equals 44.

If students struggle to express themselves with written responses, consider providing sentence stems or answer choices. This will enable students to demonstrate their understanding.

Students may struggle to understand that all three equations are correct and can be used to solve. Encourage students to think flexibly about the equations. To further scaffold, ask questions like:

- *What does the variable s represent in this problem?*
- *Is there only one way to solve for s based on the problem?*
- *What do you notice about each of the equations?*

14.8.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

Component Overview: *Students create related multiplication and division equations to represent two real-world situations. Students then choose an equation to solve for the unknown variable and explain how the multiplication and division equations demonstrate inverse operations.*

14.8.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

For each situation, write an equation that can be used to determine the unknown value. Then, solve the equation. Be sure to check your solution.

1. It takes 5 yards of fabric to make a Native American ceremonial dress.
   a. Write a division equation and a multiplication equation that can be used to determine how many yards, d, of fabric are needed to make 17 ceremonial dresses.

Division Equation	Multiplication Equation
$\frac{d}{5} = 17$	$5 \cdot 17 = d$

   b. Solve either of the equations for d. Use the other equation to check your work. $d = 85$

2. Drake is selling candy bars for a school fundraiser. Each candy bar sells for $\$3.00$.
   a. Write a division equation and a multiplication equation that can be used to determine how many candy bars, c, he sold if he raised $\$399$.

Division Equation	Multiplication Equation
$\frac{399}{3} = c$	$3c = 399$

   b. Solve either of the equations for c. Use the other equation to check your work.
   $c = 133$

3. Discuss with your partner how creating the two different equations in Questions 1 and 2 demonstrate inverse relationships. *Answers will vary. Creating two different equations shows inverse relationships by demonstrating how the operations undo each other to result in the same answer.*

Consider allowing students to reference a multiplication table while writing and solving the equations. It may be beneficial to encourage students to create fact families to help create the equations.

Students may have trouble articulating how the equations demonstrate inverse relationships. Circulate as students discuss question 3 with their partners. If you notice students struggling, ask questions to provide further scaffolding:

- *How are the equations similar? How are they different?*
- *What makes operations inverses?*
- *How do the equations show this relationship?*

14.8.5 Try It: Mixed Review

Component Overview: *Students engage in a mixed review involving writing and solving one-step equations.*

14.8.5 Try It: Mixed Review

Select all of the equations that represent each situation.

1. One Valencia orange tree can produce 275 oranges. A crate contains 25 oranges.

   a. How many crates can one Valencia orange tree fill up? Select all that apply.

   - ☐ $\frac{x}{25} = 275$
   - ☐ $\frac{x}{275} = 25$
   - ☐ $275 = 25x$
   - ☐ $\frac{275}{x} = 25$
   - ☐ $275x = 25$
   - ☐ $\frac{275}{25} = x$

   b. Pick one of the equations you selected and solve the problem. Show your reasoning.
   Answers will vary.
   $x = 11$

2. Aliyah and Tiana want to bake 3 carrot cakes. Each carrot cake needs 2 cups of salted butter.

   a. How much butter will they need to bake all 3 cakes? Select all that apply.

   - ☐ $\frac{x}{2} = 3$
   - ☐ $6x = 2$
   - ☐ $2x = 3$
   - ☐ $\frac{x}{3} = 2$
   - ☐ $2(3) = x$
   - ☐ $\frac{x}{3} = 6$
   - ☐ $3x = 2$

   b. Select one of the equations you selected and solve the problem. Show your reasoning.
   Answers will vary.
   $x = 6$

Consider creating and displaying an anchor chart with instructions, examples, and visual models for students to reference as they complete these questions independently.

If students struggle to solve equations algebraically, provide them with concrete manipulatives or tell them to draw visual models. Additionally, if students forget to balance their equations, explain that to keep the equation balanced, any operation performed on one side of the equals sign must be performed on the other side.

3. Solve each equation for the variable. Verify your answers.

a. $5x = 230$ $\frac{5x}{5} = \frac{230}{5}$ $x = 46$	b. $y + 4 = -2$ $y + 4 - 4 = -2 - 4$ $y = -6$
c. $-16 + d = -26$ $-16 + 16 + d$ $= -26 + 16$ $d = -10$	d. $r - 96 = 103$ $r - 96 + 96 = 103 + 96$ $r = 199$
e. $\frac{k}{4} = -51$ $\frac{k}{4}(4) = -51(4)$ $k = -204$	f. $-156 = -13w$ $\frac{-156}{-13} = \frac{-13w}{-13}$ $w = 12$

Remind students to check their solutions. If students finish early, ask them to write real-world problems that can be represented by the equations.

14.8.6 Wrap-Up: Reflection

Component Overview: *Students self-assess their mastery of writing and solving one-step equations.*

14.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning objective.
   Answers will vary.

I need help. I got it. I could help someone else.

Determine the value of an unknown decimal or fraction in an equation.	
Translate a written description into an algebraic expression.	
Write a description within a real-world context given an algebraic expression.	
Write and solve one-step equations in one variable using addition and subtraction.	
Write and solve one-step equations in one variable using multiplication and division.	
Determine the value of an unknown decimal or fraction in an equation.	

Consider including an example problem for each item to help students accurately self-assess their mastery of the target.

14.9 Solving One-Step Inequalities Using Addition and Subtraction

Lesson Introduction

Benchmark	MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.	Learning Targets	• I can relate the solution set of an inequality to its graph on a number line. • I can relate the solution set of an inequality to the context it represents. • I can determine the minimum or maximum value that makes an inequality true. • I can solve one-step inequalities in one variable using addition or subtraction. • I can relate properties of inequality to solving one-step inequalities algebraically.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.	MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Essential Questions	• What is the minimum or maximum value that makes the inequality true? • How do you relate the solution set of an inequality to the context it represents? • How do you determine if solutions are reasonable for the context? • How can I use properties of inequalities to solve one-step addition and subtraction inequalities? • How do I show the solution of an inequality on a number line?
Lesson Narrative	In this lesson, students read, graph, and interpret inequalities in the form $x + p < q$. This lesson is an exploration with scaffolding where students connect algebraic representations of an inequality to a number line and a real-world context. Students also explore properties of inequality to get an understanding of the process of solving inequalities. Then students use properties of inequality to solve one-step addition and subtraction inequalities. Students also graph the solution sets and test values to prove their solution is true.

Component Summary	14.9.1 Warm-Up (~5 min.)	Identify and justify which inequality does not belong *(SE p. 384)*	14.9.4 Exploration (10 min.)	Explore properties of inequalities *(SE p. 387)*
	14.9.2 Exploration (10-15 min.)	Represent inequalities in the context of real-world problems *(SE p. 384)*	14.9.5 Guided Instruction (10 min.)	Learn how to solve and graph one-step addition and subtraction inequalities *(SE p. 388)*
	14.9.3 Bring It Together (5-10 min.)	Consider constraints on the solution sets based on the context of inequalities *(SE p. 386)*	14.9.6 Wrap-Up (~5 min.)	Reflect on learning targets by completing a Know, Wonder, and Learned table *(SE p. 390)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: • Solution set • Constraint • Addition property of inequality • Subtraction property of inequality • Asymmetric property of inequality Additional Terms: • Maximum • Minimum • Inequality	• (Optional) Number line templates • (Optional) Colored pencils • (Optional) Chart paper • (Optional) Markers	• Consider preparing sentence stems for constructive response problems.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle understanding that $x > a$ and $a < x$ are equivalent, because they are not accustomed to the variable being on the right side. • Students may struggle determining whether to shade the circle on the number line. • Students may struggle finding the maximum or minimum. • Students may struggle to understand that the value (p) for a strict inequality $(x > p$ or $x < p)$ is not the minimum or maximum. • Students may have a hard time interpreting the solution sets within the context.	• Consider creating and displaying an anchor chart and word wall for students to reference throughout the lesson. Include instructions, definitions, examples, and visual models. • Consider having students practice reading the inequalities aloud from right to left and left to right.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share In 14.9.2 Exploration, have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.	**MA.K12.MTR.7.1: Real-World Contexts** In 14.9.3 Bring It Together, students solve, graph, and interpret inequalities based on real-world situations. Students learn that sometimes solutions do not make sense in the context of a problem. For example, if determining the possible height of a person, negative solutions do not make sense.

Differentiate Instruction

Visual, auditory, and kinesthetic entry points are built in throughout this lesson. Number lines can be physical or digital manipulatives that benefit students who learn best with hands-on resources.

Provide English Language Learners with anchors of support, such as vocabulary cards of the words used in this lesson: minimum, maximum, inequality, solution set, constraint.

Lesson Notes:

14.9.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students are given four different inequalities and asked to determine which inequality does not belong and explain why. Having students raise their hands when they have a solution can discourage other students from attempting the problem. To encourage all students to participate, ask students to hold their hands over their hearts and lift a finger to signify how many solutions they have determined.*

14.9.1 Warm-Up: Which One Doesn't Belong?

Four inequalities are shown.

Inequality A	Inequality B	Inequality C	Inequality D
$x \geq 5$	$12 \leq 7$	$2 > x$	$x > 1$

1. Javier says that Inequality B does not belong with the others. What is one reason Javier might have said this?
   The inequality B is wrong because 12 is greater than 7.

2. Select a different inequality and explain why it doesn't belong.
   Answers will vary.
   $2 > x$
   I chose this one because in all the other equations x is greater.

Students may need a refresher on how to read inequalities. Model how to read inequalities from left to right and from right to left. Then have students practice reading the inequalities themselves.

After initially identifying one inequality that does not belong, challenge students to think of as many reasons as they can for each inequality not belonging.

14.9.2 Exploration: Representing Inequalities in Context

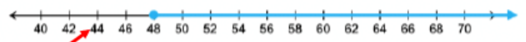
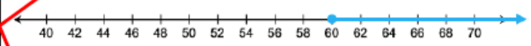
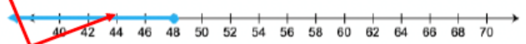
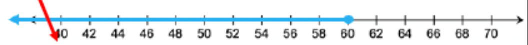
Component Overview: *Students explore how to represent inequalities in context. Students initially match height restrictions of attractions with number lines. Next, they use substitution to determine whether values are contained in the solution sets of inequalities. Finally, students write, graph, and determine the minimum solution of an inequality that represents the height restrictions of a ride.*

14.9.2 Exploration: Representing Inequalities in Context

Height restrictions are often in place for roller coasters and rides at amusement parks to keep guests safe. At Disney's Typhoon Lagoon Water Park, the following height restrictions are in place.

Attraction	Height Restriction
Bay Slides	Guests must be 60 inches or shorter
Crush 'n' Gusher	Guests must be 48 inches or taller
Ketchakiddee Creek	Guests must be 48 inches or shorter

1. Work with your partner to determine which graph represents the height restrictions of each attraction at Typhoon Lagoon. Then, draw an arrow from each attraction to its matching graph.

Attraction	Graph Representing Safe Heights for the Attraction
Bay Slides	
Crush 'n' Gusher	
Ketchakiddee Creek	
	

2. Nick is happy to realize that he is tall enough to go on the Crush 'n' Gusher raft ride. Nick's friend Matt is 2 inches shorter than Nick. Explain whether Matt is tall enough to go on the ride with Nick.
Answers will vary. If Nick is 48 inches tall then Matt would not be tall enough because he would be 46 inches. However, if Nick is 50 inches or taller than Matt is tall enough because he would be at least 48 inches.

- *How do you determine from the height restriction chart whether the circle is closed or open?*
- *How do you determine from the height restriction chart which direction to shade?*
- *What values make sense in this context?*

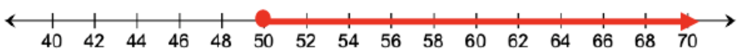
Think-Pair-Share
Have students initially complete question 2 independently. Then have students discuss their solutions and reasonings with their partners. These discussions will help students build confidence and understanding.

3. Complete the statements.

   If Nick is n inches tall, then Matt is [○ $n+2$ / ○ $n-2$] inches tall.

   For Matt to be tall enough to go on Crush 'n' Gusher, the value of [○ $n+2$ / ○ $n-2$] must be [○ less than / ○ greater than] or equal to [○ $48.$ / ○ $60.$] Therefore, the solutions to the inequality [○ $n+2 \leq$ / ○ $n+2 \geq$ / ○ $n-2 \leq$ / ○ $n-2 \geq$] [○ 48 / ○ 60] represent Nick's height that mean Matt could also ride the Crush 'n' Gusher.

4. On the number line, show all the possible heights that Nick could be if Matt is also tall enough to go on the ride.



5. What is the minimum height Nick must be in order for Matt to be able to go on the ride as well?
   Nick must be at least 50 inches for Matt to go on the ride as well.

Julia noticed the graph also represents the solutions to the inequality $n \geq 50$.

6. Work with your partner to explain why the solutions to the inequality $n-2 \geq 48$ can be represented the same way on a number line as $n \geq 50$.
   Answers will vary. The solution to $n-2 \geq 48$ is the same set of numbers as the solution to $n \geq 50$, therefore the graphs will be the same.

For question 3, consider modeling Guess and Test for students. For example, "If Nick is 52 inches tall, then how tall is Matt?" Students should respond with 50 inches. This example will help students understand why the expression $n-2$ represents the context.

Question 6 is a good place to offer an extension. Ask students how they think they could solve the inequality using algebraic thinking.

14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Component Overview: *Students consider constraints on the solution sets of inequalities based on the contexts. Students also determine whether the minimum or maximum values will be included or excluded based on whether the circle is open or closed. Students reason about whether solutions make sense in the context of problems.*

14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Inequalities have an infinite number of solutions. The set of all the solutions of an inequality is called the ***solution set***.

The ***solution set*** of an inequality includes the values that, when substituted for the variable, make the inequality true.

Before moving, check for understanding regarding the definition of solution set. Ask students what "an infinite number of solutions" means.

One way to represent the solution set of an inequality is by shading the values on the number line, beginning with the minimum or maximum value, and including all the values to the right or left of it that are included in the solution set.

1. Complete each statement.

   A minimum or maximum value represented by an open circle means the value is
   ○ included in
   ○ excluded from
   the solution set.

   A minimum or maximum value represented by a closed circle means the value is
   ○ included in
   ○ excluded from
   the solution set.

Graphs representing the solution set of inequalities are helpful when interpreting them in real-world contexts. However, sometimes there are parts of the solution set that may not make sense in a given context.

14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts (continued)

For example, to ride the Bay Slides at Typhoon Lagoon, guests must be 60 inches or shorter. The graph below represents the graph of $h \leq 60$ in this situation, where h is the height of guests that can ride the Bay Slides.

2. What does 60 represent in this situation?
   60 represents the maximum height of guests that can go on Bay Slides.

3. What does the arrow pointing to the left on the number line represent?
   The arrow pointing towards left represent the heights of guests than can ride on Bay Slides.

4. Describe at least two values that are part of the solution set of $h \leq 60$ that do not make sense in this context.

   Answers will vary. $h = -2$ or $h = -0.25$ because you cannot have negative heights.

> Sometimes there are values that are part of the mathematical solution set that do not make sense in the context. In those cases, the variable may have a **constraint** (restriction or limitation) placed on it so that the mathematical model is realistic.

The Exploration demonstrated the inequalities $n - 2 \geq 48$ and $n \geq 50$ are equivalent because all of the same values of n make both inequalities true. Both inequalities were also represented using the same graph.

Inequalities that have the same solution set are called equivalent inequalities.

- *What is the minimum height you would expect to see on a water slide?*
- *What heights are too low?*
- *Are negative solutions reasonable in the context of this question? Why or why not?*

Students may struggle to comprehend the word "constraint." Consider using a Vocabulary Improvement Strategy to connect the meaning of the word to other contexts and a visual representation of what it means to the student.

14.9.4 Exploration: Properties of Inequality

Component Overview: *Students explore how the asymmetric property of inequalities allows them to reverse the order of the inequality. Students also explore how the addition and subtraction properties of inequality allow them to add or subtract the same number to both sides of the inequality.*

14.9.4 Exploration: Properties of Inequality

Three inequalities are shown.

$x > {}^{-}2.3$ $\qquad$ ${}^{-}2.3 > x$ $\qquad$ ${}^{-}2.3 < x$

1. Circle the inequalities that represent the same solution set.

2. Complete the statement.

   For both inequalities, all the values of x that are greater than -2.3 make both inequalities true.

These two inequalities are equivalent by the ***asymmetric property of inequality***.

The ***asymmetric property of inequality*** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \le \frac{7}{10}$.

$$\frac{7}{10} \ge q + \frac{2}{5}$$

Consider having students practice reading inequalities aloud from right to left and left to right.

The Exploration is intentionally scaffolded to make connections between solving equations and inequalities. The Exploration also is scaffolded to introduce the properties of inequality that students apply when solving addition and subtraction inequalities. While circulating, support students who may need help.

14.9.4 Exploration: Properties of Inequality (continued)

4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides	$2.1 < 6.1$	*yes*
$-5 < \frac{3}{4}$	Subtract 2 from both sides	$-7 < -1\frac{1}{4}$	*yes*
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides	$10.65 > 10\frac{1}{5}$	*yes*
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides	$-\frac{5}{6} > -1\frac{1}{2}$	*yes*

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, … *the new inequality is also true.*
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, … *the new inequality is also true.*

Students should conclude that adding or subtracting the same value to both sides of an inequality results in a true statement. Provide a visual representation of why this is true by moving the two values in the inequality on the number line the same distance.

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction

Component Overview: *Students begin by completing a table for a given expression. Using the table and the relationship of the expression to the number -2, the boundary point, solutions, and graph of the relationship is determined. Then students learn how to solve inequalities using the addition and subtraction properties of inequality.*

14.9.5 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

1. Complete the table to determine the value of $x - 3$ for each value of x.

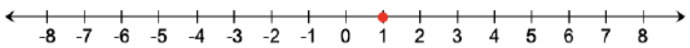
x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x-3$	-7	-6.2	-5	$-3\frac{1}{2}$	-3	-2	-1	$\frac{6}{7}$	1

Connect the answer to question 2 to the table of values.

2. For which value of x is the equation $x - 3 = -2$ true?

1

3. Represent the solution to the equation $x - 3 = -2$ on the number line.



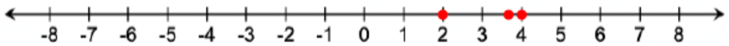
Compare and Connect
Highlight the connection between the equation and the inequality by considering the following questions.

- *What changes when the equality symbol changes to an inequality symbol?*
- *How do the graphs of the solutions compare?*

4. For which values of x in the table is the inequality $x - 3 > -2$ true?

$2; 3\frac{6}{7}; 4$

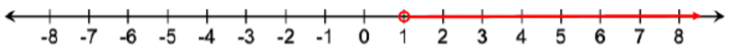
5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.



Highlight the difference between the graph of the solution to an equation vs an inequality. Students do not often graph solutions to an equation on a number line. However, seeing the difference will provide a visual entry point.

The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.

6. Graph the solution set for the inequality.



Students may answer $x > 2$ since that is the place where the value of $x - 3$ first surpasses the number -2. Remind students that there are values between 1 and 2. Ask whether $1.1 - 3 > -2$.

7. Write a value on the line to show an equivalent inequality represented by the graph above.

$x > \underline{1}$

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

14.9.5 Guided Instruction: Solving One-Step Inequalities With Addition and Subtraction (continued)

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x?
   3

Adding the same value to both sides of an inequality can be used to solve an inequality by the ***addition property of inequality***.

Addition Property of Inequality

If $a > b$, then $a + c > b + c$.

The ***addition property of inequality*** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The ***subtraction property of inequality*** can also be used to solve an inequality.

Subtraction Property of Inequality

If $a > b$, then $a - c > b - c$.

The ***subtraction property of inequality*** states that if the same value is subtracted from both sides of an inequality, the result will be an *equivalent* inequality that is also *true*.

These properties can be used to solve one-step inequalities algebraically.

Students may benefit from first using concrete values instead of abstract variables. Consider having students substitute different integers for a, b, and c for a better understanding of addition and subtraction properties of inequality.

To expand students' thinking, have them discuss what other properties closely relate to these two properties of inequality. Students should be able to relate the following properties.

- *The addition property of equality and the addition property of inequality.*
- *The subtraction property of equality and the subtraction property of inequality.*

Prompt students to anticipate any other properties previously learned that might relate to inequalities.

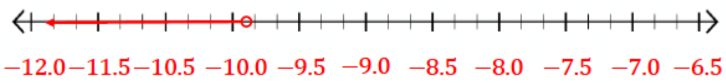
9. Use the inequality $-12.25 > y - 2.4$ to complete the following.

   a. Find the solution set by isolating the variable. Show your work.

   $$\begin{aligned} -12.25 &> y - 2.4 \\ \underline{+2.4} &\quad \underline{+2.4} \\ -9.85 &> y + 0 \\ -9.85 &> y \end{aligned}$$

   $y < -9.85$

   b. Graph the solution set on the number line.



   c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

   Answers will vary.

   $-12.25 > -10 - 2.4$
   $-12.25 > -12.4$

   $-12.25 > -12 - 2.4$
   $-12.25 > -14.4$

Now that students have been formally introduced to the addition and subtraction properties of inequality, they should use them when solving inequalities. Ask students to compare the efficiency of algebraic methods compared to using substitution and Guess and Test.

Highlight the importance of checking answers by testing values from the graph in the inequality.

14.9.6 Wrap-Up: Cool Down

Component Overview: *Students reflect on their mastery of learning targets by completing a Know, Wonder, and Learned (K-W-L chart) table. Consider using data from this formative assessment to determine which students need remediation before the next lesson.*

14.9.6 Wrap-Up: Cool Down

1. Complete the table.
   Answers will vary.

The terms and information used in today's lesson that I knew were...	Something I wondered during today's lesson was...	Something I learned during today's lesson was...

For students who struggle to represent their conceptual understanding through writing, consider providing alternative ways to assess learning for this lesson. Students could use an audio recorder or speech-to-text software, draw and label a diagram, or provide examples. All forms show an equal level of rigor and conceptual understanding.

14.10 Solving One-Step Inequalities Using Multiplication and Division

Lesson Introduction

Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Learning Targets

- I can determine the maximum or minimum value that makes an inequality true.
- I can explore the relationship between coefficients of the terms in an inequality and solution set.
- I can solve one-step inequalities in one variable using multiplication or division.
- I can relate properties of inequality to solving one-step inequalities algebraically.

Benchmark Coherence

Building On	Working Toward
MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.	MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Essential Questions

- What is the minimum or maximum value that makes the inequality true?
- How do you relate the solution set of an inequality to the context it represents?
- How do you determine if solutions are reasonable for the context?
- What is the effect multiplying and dividing by a negative number has on the inequality?
- How do I use properties of inequalities to solve one-step multiplication and division inequalities?

Lesson Narrative

In this lesson, students read, graph, and interpret inequalities in the form $px < q$ and $\frac{x}{p} < q$. Students explore the effect multiplying and dividing an inequality has on the inequality. Students explore the meaning of a minimum and maximum value in terms of inequalities and connect these to a real-world context. Students use properties of inequality to solve one-step multiplication and division inequalities. Students also graph the solution sets and test values to prove their solutions are true.

Component Summary

14.10.1 Warm-Up (~5 min.)	Write and interpret inequalities based on hanger diagrams *(SE p. 391)*	14.10.4 Collaboration (10 min.)	Explore the effect of a negative coefficient on an inequality *(SE p. 394)*
14.10.2 Exploration (15 min.)	Explore maximum and minimum values of inequalities *(SE p. 392)*	14.10.5 Bring It Together (5 min.)	Summarize the effect of a negative coefficient on an inequality *(SE p. 395)*
14.10.3 Bring It Together (5 min.)	Relate the solution set to the context it represents *(SE p. 394)*	14.10.6 Guided Instruction (10 min.)	Solve one-step inequalities with multiplication and division *(SE p. 396)*
14.10.7 Wrap-Up (~5 min.)	Compare the process of solving inequalities with positive and negative coefficients *(SE p. 397)*		

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> • Mathematical models • Multiplication property of inequality • Division property of inequality <u>Additional Terms:</u> • Solution set • Minimum • Maximum	• (Optional) Calculators • (Optional) Chart paper • (Optional) Markers	• Consider creating an anchor chart with instructions, examples, and visuals for solving one-step inequalities with multiplication and division. • Consider editing the real-world problems based on the interests of your students.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may struggle to read and interpret inequalities with the variable on the right side. • Students may not realize that $-x$ has a coefficient of -1. • Students may struggle to interpret the solution set in the context of the problem.	• Consider creating and displaying an anchor chart with vocabulary definitions, instructions, examples, and visuals. • Consider asking students to practice reading inequalities from right to left and left to right. • Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Use 14.10.4 Collaboration question 1 as an opportunity for students to initially try the problems independently and then discuss their strategies and answers with their partner. These discussions will help students gain confidence and understanding.	**MA.K12.MTR.6.1: Reasonableness** In 14.10.2 Exploration, when determining the solution set for an inequality, ask students to consider if the solution makes sense in the context they represent. Students discover that within a real-world context, often a negative value does not make sense.
Differentiate Instruction	• *For the 14.10.7 Wrap-Up, students are asked to write an explanation of the similarities and differences between solving inequalities with positive and negative coefficients. If students struggle expressing themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.*	

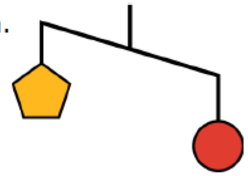
Lesson Notes:

14.10.1 Warm-Up: Visual Models

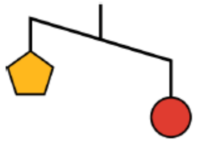
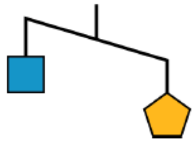
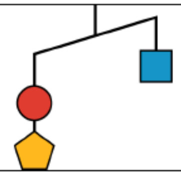
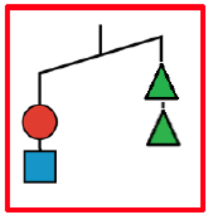
Component Overview: *Students write and interpret inequalities based on hanger diagrams. Consider first modeling how to write the description and inequality. Then ask students to work independently or with a partner. Explain to students that $p, c, s,$ and t represent the weights of the pentagon, circle, square, and triangle, respectively.*

14.10.1 Warm-Up: Visual Models

An unbalanced hanger diagram is shown. It shows that the weight of one circle is greater than the weight of one pentagon.



1. Complete the table.

Hanger Diagram	Description	Inequality
	The weight of one circle, c, is greater than the weight of one pentagon, p.	$p < c$
	The weight of one pentagon, p, is greater than the weight of one square, s.	$s < p$
	The weight of one square, s, is less than the weight of a circle and a pentagon.	$c + p > s$
	The weight of one square, s plus the weight of one circle, c is greater than the weight of two triangles, t.	$s + c > 2t$

As students complete the Warm-Up, circulate the room, listening for student misconceptions. Consider creating a chart to make notes that may inform remediation.

- *Is there another way you could write the inequality $p < c$?*
- *Can you read the inequality from right to left?*

14.10.2 Exploration: Determining the Minimum or Maximum Value That Makes an Inequality True

Component Overview: *Students explore how to determine the maximum and minimum values of inequalities. Students should use substitution to determine if a value will satisfy the inequality. Students also determine which solutions make sense in the context of the problem.*

14.10.2 Exploration: Determining the Minimum or Maximum Value that Makes an Inequality True

Work with your partner to complete the following.

1. Bak Middle School of the Arts in West Palm Beach, FL is hosting a rehearsal for its upcoming musical. The stage manager has $\$83$ from a recent fundraiser that can be used to order lunch for the cast and crew. The school's cafeteria can make boxed lunches for a cost of $\$6$ each. The inequality $6n \leq 83$ represents the situation, where n represents the number of boxed lunches that can be ordered from the cafeteria.

   a. Select the values that represent a valid number of boxed lunches the stage manager can order from the cafeteria.

   - ☐ 14
   - ☐ $13\frac{5}{6}$
   - ☐ 12
   - ☐ 9.5
   - ☐ 2
   - ☐ $^{-}4$

   b. Explain why some values in the previous question are not valid in this context.
   14 is too large of a number and -4 does not make sense in this context.

   c. The mathematical solution to the inequality $6n \leq 83$ is $n \leq 13\frac{5}{6}$. What is the maximum number of boxed lunches the stage manager can order from the cafeteria?
   13 is the maximum number because the cafeteria may not sell $\frac{5}{6}$ a lunch.

   d. What is the minimum number of boxed lunches the stage manager can order from the cafeteria?
   The least the stage manager can order is 0 boxed lunches.

   e. The graph of the solution set for $6n \leq 83$ is shown.

Remind students that they can select multiple values for question 1a.

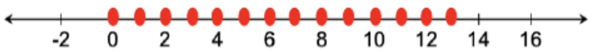
Encourage students to substitute each value into the inequality to test if it is true. For question 1b, consider highlighting the fact that -4 would not make sense in the context of the problem.

- *What is the difference between maximum and minimum?*
- *How can you determine the maximum in this situation?*
- *How can you determine the minimum?*
- *For question 1c, should you round up or down? Why?*

14.10.2 Exploration: Determining the Minimum or Maximum Value That Makes an Inequality True (continued)

Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line shown.

This is not a good representation because you cannot buy a negative number of lunch boxes.



2. Hiromi walks around his neighborhood each afternoon, tracking his pace each day. The fastest rate Hiromi can walk for an extended period of time is 3.7 miles per hour.

   a. What is the maximum distance Hiromi can walk in 0.75 hours?

   2.775 miles

   b. The inequality $\frac{d}{3.7} \le 0.75$ can be used to represent the distance Hiromi walks, d, in 0.75 hours given his fastest walking rate of 3.7 miles per hour.

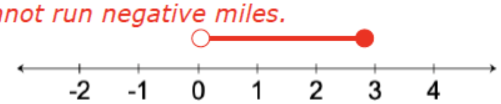
   Use your answer to Part A to determine the solution set for $\frac{d}{3.7} \le 0.75$.

   $d \le 2.775$ miles

   c. The graph of the solution set for $\frac{d}{3.7} \le 0.75$ is shown.

   Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line.

   This is not a good representation because you cannot run negative miles.



Consider providing students with the following fill-in-the-blank number sentence:

______ × ______ = ______.

- *If the maximum distance Hiromi can walk is 2.775 miles, how can we write that using an inequality?*
- *If 2.775 is the maximum, which way should the inequality symbol be facing?*
- *Should 2.775 be included or not included in the solution?*

Think-Pair-Share
For question 2c, have students think independently first before sharing with a partner. Ask students to consider whether negative numbers are possible in this context.

14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Component Overview: *Students relate solution sets to the contexts they represent. Ask students to look back at the questions from 14.10.2 Exploration and consider if any values should be excluded based on the context of the problem.*

14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Each of the inequalities in the Exploration has a mathematical solution set that can be represented on a number line. These graphs are examples of mathematical models.

Mathematical models are helpful tools used to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context. When interpreting models to solve problems, it is important to analyze whether aspects of the model make sense in a given context.

Consider utilizing some time in this section to address misconceptions that students had during the previous 14.10.2 Exploration.

14.10.4 Collaboration: The Effect of a Negative Coefficient

Component Overview: *Students explore and discuss the effects of negative coefficients on inequality symbols by comparing inequalities with positive and negative coefficients. Remind students that they can always see a visual representation of the inequality by graphing it on a number line.*

14.10.4 Collaboration: The Effect of a Negative Coefficient

1. Two simple inequalities are shown.

$$1 > 0 \text{ and } \cancel{{}^{-}1 > 0}$$

   a. Circle the inequality that is true and cross out the inequality that is false.
   b. Complete the statement.
   If 1 is greater than 0, then the opposite of 1 is ____*less*____ than 0.
   c. Write the correct inequalities by filling in a comparison symbol on each line.
   $0 < 1$ and $0 > -1$
   d. Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.

2. Work with your partner to complete the following.
   a. Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	$0 > q$
$0 > -m$	$0 < m$
$-w \leq 0$	$w \geq 0$

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

   b. Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.
   Multiplication or division by -1
   c. Describe what happens to the comparison symbol in the inequality when either of these operations are applied.
   The sign flips or changes to the opposite direction.

Prompt students to practice reading inequalities from right to left and left to right. This practice will help students who have a hard time reading or interpreting inequalities with the variables on the right.

Think-Pair-Share
Use this routine for question 1d. Ask students to justify their answers to their partners. These discussions will build student confidence and understanding.

Why is it possible that both multiplying by -1 to both sides and dividing by -1 on both sides result in the same solution?

14.10.5 Bring It Together: The Effect of a Negative Coefficient

Component Overview: *Students consolidate their understandings by summarizing the effect of a negative coefficient on an inequality.*

14.10.5 Bring It Together: The Effect of a Negative Coefficient

1. Complete the statements using words from the bank shown. Some words may not be used.

adding	subtracting	multiplying
dividing	opposite	

When the variable term in an inequality is negative, the inequality can be rewritten by ___*multiplying*___ or ___*dividing*___ the inequality by -1. In the resulting inequality, each term and the comparison symbol will change to its ___*opposite*___.

ELL students may benefit from visual representations for each word in the word bank.

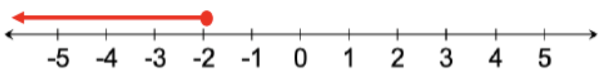
Which inequality, $-n \geq 2$ or $n \leq -2$, is easier to graph on a number line?

2. Use the inequality $-n \geq 2$ to complete the following.

   a. Complete the inequality to create an equivalent inequality.

   $$n \boxed{\leq} -2$$

   b. Graph the solution set for $-n \geq 2$.



14.10.6 Guided Instruction: Solving One-Step Inequalities With Multiplication and Division

Component Overview: *Students are guided through algorithms for solving one-step inequalities using the multiplication and division properties of inequalities.*

14.10.6 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the ***multiplication property of inequality***.

Multiplication Property of Inequality
If $a > b$ and $c > 0$, then $a \times c > b \times c$.
If $a > b$ and $c < 0$, then $a \times c < b \times c$.

1. Find the solution to both inequalities using the multiplication property of inequality.

$\frac{y}{4} \leq 2.1$	$\frac{y}{-4} \leq 2.1$
$4 \times \frac{y}{4} \leq 2.1 \times 4$	$-4 \times \frac{y}{4} \leq 2.1 \times -4$
$y \leq 8.4$	$y \geq -8.4$

Dividing both sides of an inequality by the same value can be used to solve an inequality by the ***division property of inequality***.

Division Property of Inequality
If $a > b$ and $c > 0$, then $a \div c > b \div c$.
If $a > b$ and $c < 0$, then $a \div c < b \div c$.

2. Find the solution to both inequalities using the division property of inequality.

$-25 > 5c$	$-25 > -5c$
$-\frac{25}{5} > \frac{5c}{5}$	$-\frac{25}{-5} > -\frac{5c}{-5}$
$-5 > c$	$5 < c$

Note that c cannot equal 0 for the multiplication property of inequality. That is why the property states c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (For the multiplication property, c cannot equal 0 because the two expressions would be equal, or both be 0, so the inequality statement would no longer be true.)

Consider initially presenting the properties using concrete examples (with numbers). After understanding the concrete examples, students will be ready for the abstract generalizations presented with variables.

Note that c cannot equal 0 for the division property of inequality. That is why the property states that c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (Since division by 0 is undefined, it is not possible to divide either expression by 0.)

If students forget to reverse the direction of the inequality symbol when multiplying by -5, prompt them to check their work by substituting in a value for c. This approach will empower students to self-regulate as they identify and correct their own mistakes.

14.10.7 Wrap-Up: Cool Down

Component Overview: *Students demonstrate their mastery of the lesson learning targets by explaining the similarities and differences between solving inequalities with positive and negative coefficients. Consider using data from this formative assessment to determine which students need remediation before the next lesson.*

14.10.7. Wrap-Up: Cool Down

Two inequalities are shown.

$2.5x \geq -10$ $\qquad$ $-2.5x \geq -10$

1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?
You must divide by the coefficient on both sides.	*When dividing by the coefficient in the second equation the inequality sign must change to the opposite direction.*

If students struggle to express themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.